

JANNE E. NOLAN CENTER
COUNCIL ON STRATEGIC RISKS

JULY 2025

ASSESSING DUAL-USE ISSUES AT THE AIxBIO CONVERGENCE

FUTURE RISKS AND OPPORTUNITIES FOR THE UNITED STATES

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CONVERGING RISKS LAB
COUNCIL ON STRATEGIC RISKS

Assessing Dual-Use Issues at the AIxBio Convergence:
Future Risks and Opportunities for the United States

July 2025

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Table of Contents

Executive Summary 1

Introduction 7

Scope 10

- Biotechnology 10
- Artificial Intelligence (AI) 11
- Pinpointing Uplift 13

I. The AIxBio Landscape 19

The Technologies 21

- Mapping AI-Enabled Technologies' Impacts on the Biological Weapon Development Risk Chain 21
- General-purpose AI: Foundation Models 24
- Narrow AI: AI-powered biological tools 32
- Applied-AI: the Digital-Physical Frontier 35

The Actors 39

- Competitors and Adversaries 41

Risk Mitigation Responses 48

- Multilateral Initiatives 48
- U.S. Government Actions 54
- International Responses 59

II. Methodology 69

- Initial Deepdive 69
- Plausible Futures: Scenario Planning and Backcasting Exercises 70

III. Policy Recommendations for the US Government 83

Executive Summary

Emerging and disruptive technologies, including artificial intelligence and biotechnology, have the potential to influence the threat landscape and battlespace significantly. Such technologies, both individually and as they converge, simultaneously offer both new military and civilian solutions to historic problems and generate potential threats from adversarial states and non-state actors. The convergence of AI and biology has been facilitated by the digitization of biology, which has gradually brought physical experimentation with living organisms into the cyber domain. AI technologies are now enabling the faster design of effective vaccines, therapeutics, and diagnostics, thereby expediting the delivery of medical countermeasures and enhancing the detection of threats in previously intractable datasets.

This evolution has created specific dual-use risks and accelerated others. In response, CSR initiated a study to advance the intellectual, stakeholder, and policy foundations necessary to anticipate and address the risks that arise at this nexus. This study equips policymakers to make more informed decisions as they consider how to balance concerns related to innovation, security, and safety. Yet, as this report demonstrates, the field is highly dynamic—a feature that will persist for the foreseeable future.

To more effectively understand and anticipate the opportunities and risks arising at this nexus, building on in-depth research and expert interviews, CSR employed methods within the strategic foresight toolkit. This included scenario exercises, horizon scanning, and backcasting to develop a more nuanced understanding of how future uncertainties might unfold, stretch, and stress-test expert assumptions, and conceptualize global complexities that could enable or constrain alternative future pathways. Through scenario mapping with a diverse

cohort of experts from academia, governments, industry, and think tanks, a preferred future scenario was articulated for the AIxBio nexus and used to identify the steps that could be taken over the next five years to achieve that preferred scenario.

AIxBIO IN 2030: THE "PREFERRED" SCENARIO

Great power competition persists, yet world powers increasingly forgo competition for cooperation. Russia and Ukraine come to an agreement resolving the war in Ukraine. Russia focuses on rebuilding its economy, including global economic ties. NATO maintains the status quo and does not threaten to further expand along Russia's periphery. China and Taiwan experience a detente, and China is optimistic about eventually re-integrating Taiwan in the long term through negotiation. US-China relations are warming, leading to increasingly intertwined economies and cooperation in other areas. The United States maintains a strong military deterrent in the region, disincentivizing Chinese military action against Taiwan. Multilateral organizations play a mediating role in successfully stabilizing these conflicts, thereby improving the UN's reputation and enhancing its soft power. The US, China, and Russia increasingly view the UN as an effective mechanism for resolving disagreements.

By 2030, governments will have catalyzed sustained and responsible AI development through sophisticated regulation and strong public-private collaboration. A global social consensus emerges that recognizes the need for responsible innovation in AI. This contributes to the UN becoming a productive and respected forum for establishing international governance of new AI capabilities. Governments share a nuanced understanding of AI, viewing it as a strategic capability that can deliver significant competitive advantages, yet they are clear-eyed about the downstream risks. Governments seek to regulate their domestic AI industries in line with UN agreements, while also supporting AI companies as economic 'national champions'. Frontier laboratories in the US, UK, EU, and PRC share a common understanding of the risks associated with AI, particularly in biosecurity, cybersecurity, and the integration of AI into safety-critical systems. A strong culture of public-private collaboration emerges with industry leaders closely involved in UN efforts to regulate these areas internationally. This leads to avant-garde international agreements that allow governments to

develop streamlined regulation that broadly incentivises responsible innovation.

These factors facilitate responsible innovation, championed as AI-enabled biotechnology development accelerates. Although AI-first companies initially adopt new AI models, biotechnology companies also leverage AI to accelerate their research, development, and deployment, leading to rapid growth in the sector. A social consensus emerges, recognizing AI-enabled biotechnology as a public good that is revolutionizing drug discovery, improving global public health outcomes, and reducing the fiscal burden of healthcare. AI-enabled biotechnology companies receive additional support from state sponsors, which come with regulatory conditions that serve as effective guardrails, incentivizing responsible innovation rather than stymying development.

A backcasting exercise conducted with our group of AIxBio experts and follow-up engagement with key stakeholders yielded the following practical framework and tangible recommendations for the US government to achieve the best-case scenario for societal resilience to AIxBio risks over the next five years:

1. STRENGTHENING THE BIOSECURITY AND BIOSAFETY FOUNDATION

With AI serving as a threat multiplier to existing biorisks in both quantity and quality, the development and effective implementation of strong biosecurity and biosafety policies have become even more critical. These include:

- advancing an even more comprehensive deterrence-by-denial strategy (by establishing increased national and international preparedness, early warning systems, and response capabilities),
- sufficiently funding targeted capability development;

- maintaining the relevant subject-matter expertise;
- investing in close collaboration with allied and partner countries;
- establishing and implementing mandatory DNA synthesis screening;
- improving data security frameworks;
- frameworks for controlling and monitoring access to biological data with dual-use potential;
- and updating and tailoring biosafety frameworks.

2. AIxBIO POLICY

The US government's new AI Action Plan, released on July 23, 2025, reinforces CAISI's role in evaluating frontier AI systems for national security risks, including bio risks. Beyond establishing voluntary agreements with private sector AI developers and evaluators and leading unclassified evaluations of AI capabilities, the US government needs to develop a national regulatory framework to avoid a convoluted regulatory landscape. Additionally, the US government should:

- leverage flexible hiring authorities and contracting vehicles;
- conduct classified evaluations;
- develop guidance for dual-use relevant AIxBio research; and
- establish industry-intelligence community back channels.

3. AIxBIO TECHNICAL GOVERNANCE

The rapid technical developments and applications at the AIxBio intersection require sustained investment in qualitatively updating and quantitatively universalizing the implementation of such standards—for both foundation models and AI-powered biological tools. The US government should require AI developers and, where applicable, downstream actors to uphold industry best practices, which currently include:

- Identifying, measuring, mitigating, monitoring, and responding to potential misuse risk and transparently disclosing relevant practices;
- conducting comprehensive evaluations such as expert-led red teaming and human uplift studies;
- combining relative capability, bottleneck, and threat simulation assessments for biosecurity;
- evaluating the capabilities of biological AI models and monitoring the use and security practices of autonomous labs;
- establishing data standards for biological data and monitoring the types of data being collected and aggregated; and
- leveraging AI for accelerating the development, manufacturing, and distribution of biological countermeasures.

This framework of recommendations can help guide policymakers as they prioritize and learn to navigate the AIxBio tightrope—balancing innovation with security & safety.



Source: Adobe Stock

Introduction

Emerging and disruptive technologies, including artificial intelligence and biotechnology, have the potential to influence the threat landscape and battlespace significantly. Such technologies, both individually and as they converge, simultaneously offer both new military and civilian solutions to historic problems and generate potential threats from adversarial states and non-state actors. While this revolution has helped with specific present-day issues, potentially game-changing technologies and techniques also present risks: risks that the very capabilities used to improve humanity can also be used to harm it. This conundrum is known in the life sciences and other disciplines as the dual-use dilemma.[1, 2]

The past 60 years have witnessed unprecedented advancements in the life sciences and biotechnology, and many have extrapolated these trajectories and recognized that the 21st century will be the century of biology.[3] Simultaneously, concerns have grown more significant that nation-states, groups, and individuals are increasingly able to exploit the fruits of life sciences research for nefarious purposes. Indeed, a 2004 National Academies report known as "The Fink Report" confirmed many of the developments that could augment the abilities of nefarious actors seeking to misuse biotechnology advances: activities such as developing or altering pathogens to evade existing diagnostics and

1 Epstein, G (2011). "Preventing Biological Weapons Development Through the Governance of Life Science Research," *Biosecurity and Bioterrorism*, Vol. 10, No. 1: pp. 17 - 37.

2 The UK AI Security Institute defines dual-use science as"research and technology that can be applied for beneficial purposes, such as medicine or environmental solutions, but also potentially misused to cause harm, such as in biological or chemical weapon development." Bengio, Y (2025). International Scientific Report on the Safety of Advanced AI: Interim Report. Available: https://www.gov.uk/government/publications/international-ai-safety-report-2025/international-ai-safety-report-2025

3 VENTER, C. and COHEN, D. (2004), The Century of Biology. New Perspectives Quarterly, 21: 73-77. https://doi.org/10.1111/j.1540-5842.2004.00701.x

medical countermeasures, increase the host range of a pathogen, and even potentially make diseases more severe.[4]

AIxBio convergence has been facilitated mainly by the "digitization of biology," which has gradually brought physical experimentation with living organisms into the cyber domain. This process began several decades ago with the dawn of the computer age, which led technologists to increasingly develop and utilize advanced techniques and software (e.g., bioinformatics) to perform scientific modeling and conduct complex analyses in the life sciences. This evolution has created specific dual-use risks and accelerated others. For example, the 2023 US Biodefense Posture Review explicitly set out the opportunities and perils that exist at the convergence of genomic modification and artificial intelligence capabilities: "The same biological and chemical science advancements created to develop life-saving medical countermeasures could also be used by potential adversaries to develop new or enhanced agents…and increase the number of proliferation pathways available to malign actors."[5]

In response, CSR initiated this study to help develop the intellectual, stakeholder, and policy foundations necessary to anticipate and address the risks that arise at this nexus. The study was designed to take a considered approach to the emerging risk nexus, carefully extrapolating potential trajectories and rigorously testing possible future scenarios with an interdisciplinary set of experts. It was envisioned that this in-depth landscape research and analysis of the key risks and opportunities arising from the AIxBio convergence would provide policymakers with the knowledge and information necessary to make

4 National Research Council of the National Academies (2004). *Biotechnology Research in an Age of Terrorism*. Washington, DC: National Academies Press.

5 U.S. Department of Defense (2023). Biodefense Posture Review, p. 7. Available: https://media.defense.gov/2023/Aug/17/2003282337/-1/-1/1/2023_biodefense_posture_review.pdf

better, more informed decisions as they consider balancing concerns related to innovation, security, and safety.

This report consists of three sections. Part one provides an exposition of the AIxBio risk landscape in early 2025, with a particular emphasis on the trajectories of change that exist at this complex intersection. Part two distills this knowledge and diverse perspectives to generate three potential futures, which will help identify new areas of competitive advantage for the United States and its allies on the world stage, as well as changes to national security and other health-related areas, and new ways to counter emergent threats. Part three presents a series of recommendations for various stakeholders, developed through consultations with key subject matter experts across academia, industry, and government.

Scope

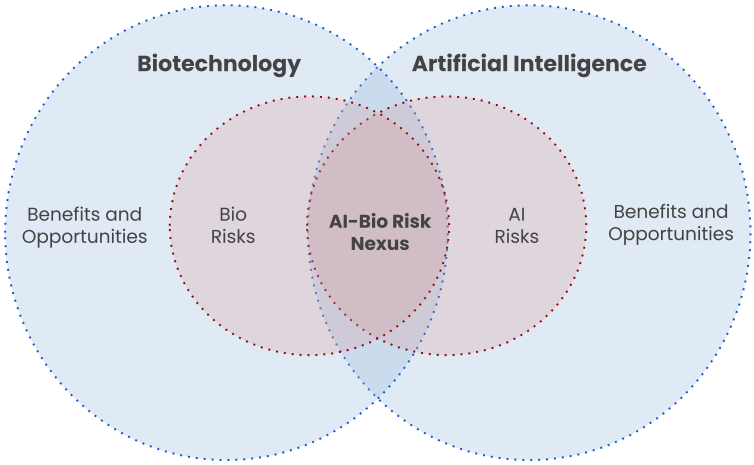


Figure 1: Visualizing the AixBio risk nexus

Biotechnology

Bioengineering in the 2010s was characterized by the convergence of AI with complementary bioengineering technologies, exemplified by DeepMind's AlphaFold system, which predicts protein structures with remarkable accuracy. Next-generation sequencing, which allows parallel processing of DNA or RNA samples, has made sequencing significantly cheaper.[6] On the applied side, the use of mRNA technology in COVID-19 vaccines has demonstrated the promise of programmable gene therapies. Cumulatively, these innovations have transformed life sciences from qualitative models to precise, quantitative control of cellular processes.

6 Siegel, V (2023). "The Human Genome Project Turns 20: Here's How it Altered the World," MIT Department of Biology, April 14, 2023. https://biology.mit.edu/the-human-genome-project-turns-20-heres-how-it-altered-the-world/

This unprecedented level of life scientific advancement will only continue to grow as these capabilities converge and interact with other rapidly emerging and disruptive technologies. As the biology of life is being explored and demystified, vast amounts of data are being generated and operationalized in combination with unprecedented computing power. Experts suggest that this is the dawn of the Fourth Industrial Revolution: a revolution that builds on the foundation of computing power and blurs "the lines between physical, digital, and biological spheres."[7]

Artificial Intelligence (AI)

To develop a proportionate regulatory approach, the UK Government established a useful AI typology that enables adequate understanding and proportionate responses to associated risks[8]:

1. Highly capable general-purpose AI: Advanced foundation models that can competently perform a wide variety of tasks and can be integrated into a wide variety of downstream applications. Generally, such models span from novice to expert capabilities, with some even exhibiting superhuman performance across a range of tasks.
2. Highly capable narrow AI: models that can exceed the capabilities of today's most advanced models in performing a narrow set of tasks, often within a specific field such as biology. When focused on biological engineering, these models are termed biological design tools (BDTs) and defined as "systems that are

7 World Economic Forum (2016). The Fourth Industrial Revolution: What it Means, How to Respond. Available: https://www.weforum.org/stories/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond/

8 UK Department for Science, Innovation and Technology (2023). Frontier AI: capabilities and risks. Available: https://www.gov.uk/government/publications/frontier-ai-capabilities-and-risks-discussion-paper

trained on biological data and can help design new proteins or other biological agents."[9]

3. Agentic AI or AI agents: An emerging subset of AI technologies that can competently complete tasks with multiple steps. These systems can use tools such as coding environments, the internet, and narrow AI models to complete tasks.

Advances in life sciences research and development over the past few decades have enabled remarkable progress in understanding and manipulating biological systems, which will become more transparent and malleable than ever through the advent of AI. Applied AI enhances software applications and leverages advanced machine learning (ML) to deliver high levels of accuracy and adaptability over time. Applied AI is contextualizing business models and industry processes, as well as changing the way we interact with everything around us. This capability increases as scientists and innovators develop more robust data sets and algorithms to contain and process biological data, more efficient training methods and models to generate outputs, and more intuitive and user-friendly interfaces. This has enormous implications for everything from basic research and development to accelerating clinical trials and even addressing previously incurable conditions.

9 Sandbrink, J (2023). Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools." arXiv. https://doi.org/10.48550/arXiv.2306.13952.

Pinpointing Uplift

Issues related to the confluence of the AI and bio fields have been recognized for many years. Yet, like the field of biological risks generally, public discourse on the topic includes a vast range of opinions, across the general public and experts in various fields. There are wide-ranging views on what the trajectory of AI holds generally, as one component of this discourse; for example, recently, there has been a proliferation of existential speculation about the "loss of control of autonomous AI systems" and the interaction with biological weapons.[10, 11] Additionally, multiple governments and private companies have undertaken practical work to monitor changes in biological risk dynamics and address them. For example, OpenAI and Anthropic have repeatedly released public "system cards" that outline the pre-deployment evaluations of their models' capabilities and the subsequent risk mitigation measures.[12, 13] Even still, some experts remain skeptical that AI will have a significant impact on the risk landscape.

Effective policymaking, therefore, requires stepping up beyond these trends in public discourse and carefully considering the nuances of the risks associated with AI models that interact with biotechnology. To make progress, it is imperative to understand where and how AI may accelerate biological weapons development capabilities—and where it does not. Additionally, it is essential to differentiate between existing

10 Saunders, W (2024). U.S. Senate, Subcommittee on Privacy, Technology, and the Law Oversight of AI: Insiders' Perspectives. Available: https://www.judiciary.senate.gov/committee-activity/hearings/oversight-of-ai-insiders-perspectives

11 Modei, D (2025). Anthropic C.E.O.: Don't Let A.I. Companies off the Hook. Available: https://www.nytimes.com/2025/06/05/opinion/anthropic-ceo-regulate-transparency.html?searchResultPosition=2

12 OpenAI (2024). Chat GPT4 System Card. Available: https://cdn.openai.com/papers/gpt-4-system-card.pdf

13 Anthropic (2025). System Card: Claude Opus 4 & Claude Sonnet 4. Available: https://www-cdn.anthropic.com/6be99a52cb68eb70eb9572b4cafad13df32ed995.pdf

biotechnology dual-use concerns and AI's uplift effect, defined by UK AISI as "how effectively humans can use AI systems to accomplish various tasks, compared to using existing technology."[14] This clarity, in turn, enables stakeholders to develop appropriate and proportionate risk mitigation strategies.

In terms of current capabilities, there are two key ways in which AI can influence biosecurity risks: (1) lowering barriers for non-experts to synthesize, acquire, and disseminate biological weapons and (2) raising the "ceiling" or amount of harm from biological agents.[15, 16]

Firstly, AI's role in **reducing the barriers to entry** for researching, developing, and weaponizing biological agents. The development and use of bioweapons have historically been hindered by the need for "tacit knowledge" in synthesizing biological agents. Despite the public availability of reverse genetics protocols, practical laboratory skills, such as cell culture, remain a barrier. AI tools have the potential to act as expert lab assistants to actors with lower tacit knowledge, mimicking the tutoring of a more advanced scientist, but potentially without the situational awareness or moral objection to potentially malicious work.[17]

Additionally, AI tools may also suggest alternative routes to obtaining agents that do not require them to perform tacit-knowledge-intensive wet lab experiments, but rather outsource experiments they are

14 UK AI Security Institute (2025). International AI Safety Report 2025. Available: https://www.gov.uk/government/publications/international-ai-safety-report-2025/international-ai-safety-report-2025

15 Rose, S and Nelson. C (2023). Understanding AI-Facilitated Biological Weapon Development. The Centre for Long-Term Resilience.

16 Sandbrink, J (2023). Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools." arXiv. https://doi.org/10.48550/arXiv.2306.13952.

17 Sandbrink, J (2023). Artificial Intelligence and Biological Misuse: Differentiating Risks of Language Models and Biological Design Tools." arXiv. https://doi.org/10.48550/arXiv.2306.13952.

incapable of carrying out on their own. One mechanism might be the use of laboratory robots, where foundation models also contribute to the advancement of autonomous science capabilities. Foundation models can help actors convert natural language comments into scripts for liquid-handling robots, facilitating some biological experiments.[18, 19]

Secondly, the community is concerned about AI's role in **raising the ceiling of harm**. AI tools can also facilitate the discovery and development of novel engineered bioweapons that are more dangerous or tactically useful than naturally emerging pathogens, increasing the amount of harm bioweapons can cause or making it more likely for skilled actors to use them.[20, 21] These tools might also be used to modify an agent optimized for immune evasion, evade medical countermeasures, or allow actors to find functional equivalents to dangerous agents that are currently restricted or regulated through sequence-based screening mechanisms.

18 Inagaki, Takashi, Akari Kato, Koichi Takahashi, Haruka Ozaki, and Genki N. Kanda. 2023. "LLMs Can Generate Robotic Scripts from Goal-Oriented Instructions in Biological Laboratory Automation." arXiv. https://www.doi.org/10.48550/arXiv.2304.10267.

19 O'Donoghue, Odhran, Aleksandar Shtedritski, John Ginger, Ralph Abboud, Ali Essa Ghareeb, Justin Booth, and Samuel G. Rodriques. 2023. "BioPlanner: Automatic Evaluation of LLMs on Protocol Planning in Biology." arXiv. https://doi.org/10.48550/arXiv.2310.10632.

20 Dauparas, J., I. Anishchenko, N. Bennett, H. Bai, R. J. Ragotte, L. F. Milles, B. I. M. Wicky, et al. 2022. "Robust Deep Learning Based Protein Sequence Design Using ProteinMPNN." bioRxiv. https://doi.org/10.1101/2022.06.03.494563.

21 Rose, S and Nelson. C (2023). Understanding AI-Facilitated Biological Weapon Development. The Centre for Long-Term Resilience.

This report aims to establish the intellectual, stakeholder, and policy foundations necessary to anticipate and address emerging risks at the AixBio nexus. Initially, the report provides an analysis of the AIxBio risk landscape, examining the technologies, actors, and government responses. Thereafter, the scenario planning and backcasting process, employed to improve our ability to anticipate strategic opportunities and threats, will be described. An articulation of the scenarios and the process for establishing a "preferred future" will be elucidated, uncovering novel insights into potential future inflection points and interdependencies. Thereafter, the report delivers a practical framework and tangible recommendations for the US government to achieve the best-case scenario for societal resilience to AIxBio risks over the next five years.



Source: Adobe Stock

I. The AIxBio Landscape

The convergence of AI with biotechnology is revolutionizing disease prediction, detection, and prevention. For several years, AI techniques have been increasingly incorporated into biomedical research, poised to radically transform the industry, reduce costs, and shorten development times. During the COVID-19 pandemic, numerous examples emerged of AI enabling faster responses. AI was incorporated into Pfizer's COVID-19 vaccine trial to clean the data of more than 30,000 patients.[22] In South Korea, the analysis of lateral flow assays was improved and accelerated by incorporating an innovative deep-learning architecture and verification mechanism.[23]

AI technologies are enabling the faster design of effective vaccines, therapeutics, and diagnostics, thereby expediting the delivery of medical countermeasures and enhancing the detection of threats in previously intractable datasets. Nobel award-winning computer scientist and CEO of Google DeepMind, Sir Demis Hassabis, suggested in early 2025 that with AI, the end of disease is "within reach...maybe within the next decade or so."[24] This was echoed by OpenAI CEO Sam Altman in 2025, who stated, "as this technology progresses, we will see diseases get cured at an unprecedented rate."[25]

22 Hutson, M (2024). How AI is being used to accelerate clinical trials. Nature 627, S2-S5 (2024) doi: https://doi.org/10.1038/d41586-024-00753-x

23 Lee S, Yoo YK, Han SI, Lee D, Cho SY, Park C, Lee D, Yoon DS, Lee JH (2023). Advancing diagnostic efficacy using a computer vision-assisted lateral flow assay for influenza and SARS-CoV-2 detection. Analyst. 2023 Nov 20;148(23):6001-6010. doi: 10.1039/d3an01189e. PMID: 37882491.

24 CBS News (2025). 60 Minutes: Demis Hassabis. Available: https://www.cbsnews.com/news/artificial-intelligence-google-deepmind-ceo-demis-hassabis-60-minutes-transcript/

25 Time Magazine (2025) Exclusive: AI Outsmarts Virus Experts in the Lab, Raising Biohazard Fears. Available: https://time.com/7279010/ai-virus-lab-biohazard-study/

There are indications that these grand predictions are becoming a reality. The latest foundational models (including GPT-4 and 4o) are already displaying significant performance increases in effectively assisting in dual-use research tasks. When combined, LLMs and narrower AI-enabled biological tools offer powerful efficiencies and cost reductions to complex R&D. Indeed, large biology-focused general-purpose models capable of distilling biology's complexity are imminent, and many predict a quantum leap in biological research as a result.

These developments, at the convergence of AI and biology, although revolutionary, create systemic risks that require ongoing attention. In a clear-eyed warning, Sir Demis Hassabis raises two concerns regarding AI's development trajectory, firstly, "bad actors, humans...users of these systems repurpose these systems for harmful ends. And then the second thing is the AI systems themselves as they become more autonomous and more powerful."[26]

This section presents some of the most critical aspects of the AIxBio context that policymakers must understand to inform their efforts, focusing on specific technologies, the actors involved, and responses to emerging risks. Naturally, the landscape has undergone substantial changes over the 18 months of this research project and will continue to evolve rapidly. To mitigate redundancy, this section describes the landscape in Q2 2025 and, in some instances, highlights how that landscape has evolved in just the past few years, enabling policymakers to understand the rapid pace of change in this field. It pays particular attention to the trajectories of change while avoiding supposition.

26 CBS News (2025). 60 Minutes: Demis Hassabis. Available: https://www.cbsnews.com/news/artificial-intelligence-google-deepmind-ceo-demis-hassabis-60-minutes-transcript/

The Technologies

Mapping AI-Enabled Technologies' Impacts on the Biological Weapon Development Risk Chain

Understanding where and how AI may accelerate biological weapons research, development, scaling, and deployment capabilities is essential for identifying intervention points and aiding the development of risk mitigation strategies. Recognising the need to contextualise the AIxBio uplift claims, in 2023, the Centre for Long Term Resilience (CLTR) reintroduced the idea of employing weapons development chain analysis to enable researchers and policymakers to pinpoint specific technologies and areas of potential uplift within this threat landscape.[27]

The framework maps the weapons development chain from malicious actors intending to cause harm using a biological agent to the deployment of a functional biological weapon. The framework helpfully demonstrates that although AI is not required to make biological weapons, AI-enabled tools have the potential to uplift capabilities at multiple steps in the chain. This, in turn, enables policymakers to understand the nuanced implications for specific policy areas without losing sight of the broader impact. As an iterative process, this is a crucial tool for pinpointing specific areas of AI uplift and scoping further evaluations.

27 Rose, S & Nelson, C (2023). Understanding AI-Facilitated Biological Weapon Development. The Centre for Long-Term Resilience: London UK. October 2023.

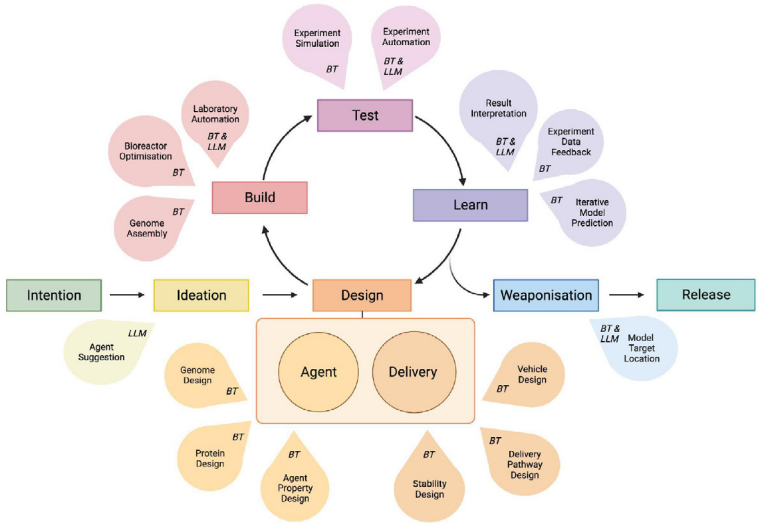


Figure 2: The Centre for Long Term Resilience's visualization of the risk chain in biological weapon development, from malicious intention to deliberate release event.[28]

28 Rose, S & Nelson, C (2023). Understanding AI-Facilitated Biological Weapon Development. The Centre for Long-Term Resilience: London UK. October 2023.

Table 1: Extrapolating CLTR's Capability Analysis Along the Biological Weapons Risks Chain[29]

Risk Chain Step	Type(s)	Influence(s) on biosecurity risks	Example of misuse
Intention	Foundation models Agentic AI	Lowering Barriers to Biological Weapons	Inspiring non-state actors to employ biological weapons to further their agenda, whether through a specifically targeted agentic AI or the actor using foundation models to develop their cause's modus operandi.
Ideation	Foundation models Biological Design Tools	Lowering Barriers to Biological Weapons Raising the "Ceiling" of Harm	State and non-state actors employ foundation models and narrow biological design tools to develop bioweapon methodologies, models, and forecasting to achieve a specific end goal.
Design	Biological Design Tools Agentic AI	Lowering Barriers to Biological Weapons Raising the "Ceiling" of Harm	State and non-state actors employ sophisticated narrow models to test, learn, and build biological weapons. Sophisticated non-state and state actors may employ agentic AI to access particularly sensitive bioresearch facilities to obtain design-relevant data for a specific bioweapon.
Weaponization	Foundation models Biological Design Tools Agentic AI Applied AI: Automated Bio-manufacturing and Cloud Labs	Lowering Barriers to Biological Weapons Raising the "Ceiling" of Harm	State and non-state actors employ a host of AI-enabled technologies to develop a bespoke and stable pathogen of specific qualities that is to be released to achieve a desired outcome. This pathogen is developed remotely via a cloud lab to avoid regulatory regimes that prohibit the purchase and use of component materials.
Release	Foundation models Agentic AI	Lowering Barriers to Biological Weapons Raising the "Ceiling" of Harm	A foundation model is employed to establish the most effective release site and methodology to achieve a specific end goal whilst mitigating blowback or other unintended consequences.

29 Rose, S & Nelson, C (2023). Understanding AI-Facilitated Biological Weapon Development. The Centre for Long-Term Resilience: London UK. October 2023.

General-purpose AI: Foundation Models

The rapid advancement of AI has far-reaching implications across multiple domains, including its potential role in the proliferation of biological weapons. Researchers have argued that the breadth of potential use cases for foundation models makes them a general-purpose technology, similar to electricity.[30] Foundation models are accelerating advances in the life sciences, from training new scientists to enabling faster scientific workflows. Significant areas of uplift provided by foundation models include:

- **Harnessing the power of bioinformatics** by improving the capacity to analyze and identify patterns in large volumes of biological data, presenting a step change in capabilities such as biological threat detection, drug discovery, outbreak prediction, and microbial forensics.
- **Knowledge mining of biomedical literature** enables the efficient extraction of information from vast amounts of text data. This process, driven by Natural Language Processing, helps researchers identify novel drug targets, accelerate drug discovery, and gain deeper insights into complex biological processes.
- **Clinical decision and diagnosis support tools** enhance the accuracy of diagnosis, optimizing treatment plans, and improving patient outcomes. Novel AI-enabled tools identify patterns and provide insights to healthcare professionals, augmenting their decision-making.
- **Workflow optimization** by generating step-by-step technical instructions for working with or even creating pathogens

30 Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2023). Gpts are gpts: An early look at the labor market impact potential of large language models. arXiv preprint arXiv:2303.10130

and toxins, and surfacing information that experts would take time to find online.

While these capabilities will have tremendous beneficial applications, there is a risk that they can be used for malicious purposes. The 2025 International Scientific Report on the Safety of Advanced AI[31] highlights a number of significant risks from malicious use of AI, including:

- **Deepfakes and the manipulation of public opinion.** Deepfake images and videos are thought to have been nefariously developed and deployed in several elections in 2024 to defame political candidates and sway results.
- **Cybersecurity risks**. Recent AI safety research indicated significant advances in cyber office-related capabilities of foundation models.
- **Bias and unintended harm from malfunctions.** Recent research suggests that foundational models have the potential to amplify existing biases, whether social, political, or otherwise. Current models are also prone to malfunction and unreliability, which could lead to unintended harm if, for example, they are used for medical advice or legal counsel.

Leading AI companies have been quick to point out the significant dual-use implications presented by the rapid development in scientific discovery, enabled by their technologies. Companies like OpenAI and Anthropic have been transparent about the biosecurity risks associated with their powerful models. They have published research on these dangers, established internal safety teams, and are actively developing and implementing safety monitoring systems to mitigate these risks. In a 2025 report, OpenAI acknowledged that "the same underlying

31 Bengio, Y (2025). International Scientific Report on the Safety of Advanced AI: Interim Report. Available: https://www.gov.uk/government/publications/international-ai-safety-report-2025/international-ai-safety-report-2025#references

capabilities driving progress, such as reasoning over biological data, predicting chemical reactions, or guiding lab experiments, could also potentially be misused to help people with minimal expertise to recreate biological threats or assist highly skilled actors in creating bioweapons. Physical access to labs and sensitive materials remains a barrier—however, those barriers are not absolute."[32]

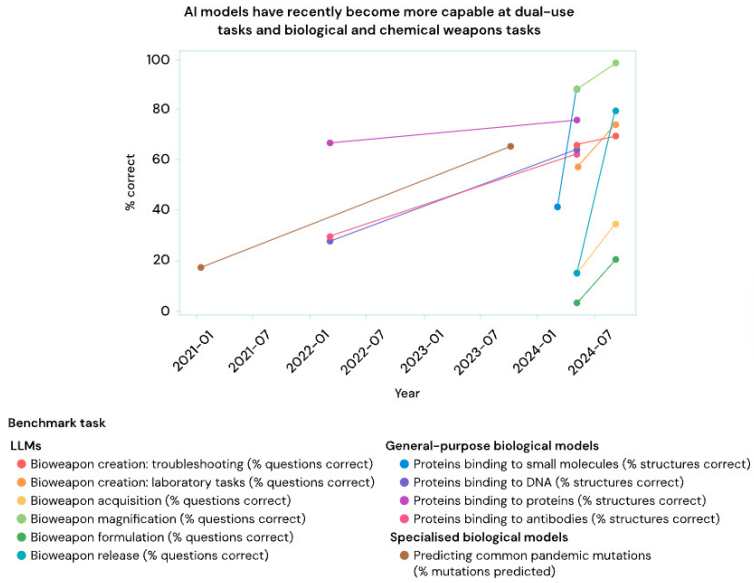


Figure 3: OpenAI's analysis of the increasing dual-use capabilities of foundational models over the past four years.[33, 34]

Understanding and measuring the capabilities of general-purpose AI is crucial for assessing its risks; however, the capabilities of general-purpose AI systems are challenging to reliably measure. The table

32 OpenAI (2025) Preparing for future AI capabilities in biology. Available: https://openai.com/index/preparing-for-future-ai-capabilities-in-biology/

33 UK AI Security Institute (2025). International AI Safety Report 2025. Available: https://www.gov.uk/government/publications/international-ai-safety-report-2025/international-ai-safety-report-2025

34 OpenAI (2024). OpenAI o1 System Card. Available: https://cdn.openai.com/o1-system-card-20240917.pdf

below provides a synopsis of eminent research (as of the time of this report's writing) aimed at measuring foundational model capabilities and deducing the subsequent dual-use implications.

Table 2: Scrutinizing Foundational Model Capabilities and Dual-Use Implications

Publication	Methodology	Findings/Comment
AIxBio Uplift Studies		
2023. Soice et al. "Can large language models democratize access to dual-use biotechnology?"[35]	Three groups of 3-4 students with no scientific training used LLMs to see how they could assist in planning and carrying out a biological attack, for example, by using them to gather information about harmful biological artifacts, or to gain guidance on how to acquire those artifacts and deploy them to cause maximum harm.	Their initial findings suggested that "LLMs will make pandemic-class agents widely accessible [...] even to people with little or no laboratory training".
2024. Rand Corporation. "The Operational Risks of AI in Large-Scale Biological Attacks."[36]	Teams of researchers, role-playing as malign non-state actors, were assigned to realistic scenarios and tasked with planning a biological attack. Some individuals had access to an LLM and the internet, while others had access only to the internet. The authors aimed to identify potential risks associated with LLM misuse, generate policy insights to mitigate these risks, and contribute to the responsible development of LLMs.	The study found no statistically significant difference in the viability of plans generated with or without LLM assistance. The authors identified problematic outputs from LLMs that increased the risk of misuse; however, the prompts did not extend beyond publicly available information on the internet. Overall, the study suggested that LLMs (in 2024) did not substantially increase the risks associated with planning biological weapon attacks.

35 Soice, E et al. (2023). Can large language models democratize access to dual-use biotechnology? Available: arxiv:https://arXiv.org/abs/2306.03809

36 Mouton, C et al. (2024) The Operational Risks of AI in Large-Scale Biological Attacks: Results of a Red-Team Study. Santa Monica, CA: RAND Corporation. Available: https://www.rand.org/pubs/research_reports/RRA2977-2.html.

Publication	Methodology	Findings/Comment
Internal Assessments (Model Cards). Red teaming assessments have been carried out by AI model developers.		
January 2024. OpenAI. "Building an Early Warning System for LLM-Aided Biological Threat Creation."[37]	The most extensive to-date human evaluation of AI's impact on biorisk information at the time, with 100 human participants. Half were biology experts with PhDs and professional wet lab experience, while the other half were student-level participants who had completed at least one university-level biology course. Each participant was assigned a set of exercises that covered various aspects of the end-to-end process for creating biological threats.	The evaluation suggested that GPT-4 provided at most a mild uplift for biologists working to create lethal viruses, and mild uplifts in accuracy and completeness for those with model access, although they were not large enough to be statistically significant.
May 2025. "Anthropic. Bioweapons acquisition uplift trial."[38]	A controlled trial measuring AI assistance in bioweapons acquisition and planning by a hypothetical adversary. Groups of 8-10 participants were asked to draft a comprehensive bioweapons acquisition plan; the control group had access to the internet, and the model group had access to Anthropic's latest LLM and research tools with safeguards removed.	Recognizing that text-based uplift trials are far weaker proxies for real-world scenarios, the trial suggested that both Clause Opus 4 and Claude Sonnet 4 provided uplift (2.53× and 1.70×, respectively, compared to the control), but neither provided significant additional risk.

37 OpenAI (2024). Building an early warning system for LLM-aided biological threat creation. Available: https://openai.com/index/building-an-early-warning-system-for-llm-aided-biological-threat-creation/

38 Anthropic (2025). System Card: Claude Opus 4 & Claude Sonnet 4. Available: https://www-cdn.anthropic.com/6be99a52cb68eb70eb9572b4cafad13df32ed995.pdf

Publication	Methodology	Findings/Comment
Late 2024. OpenAI and Los Alamos National Laboratory. "Wetlab uplift evaluations."[39]	An assessment of how frontier models could uplift human performance within laboratory settings through multimodal capabilities like vision and voice. The evaluations assessed the uplift in the ability to perform and troubleshoot a safe standard laboratory protocol.	The results of this study were not made public.
Benchmarks. Benchmarks make up around 40% of Anthropic's, 50% of OpenAI's, and 80% of Google DeepMind's biorisk evaluations.[40]		
SecureBio and the Center for AI Safety. "Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark."[41]	Measuring practical knowledge of virology, focusing on troubleshooting experiments. Three hundred twenty-two multimodal questions covering fundamental, tacit, and visual knowledge essential for practical work in virology laboratories, developed and tested by scientists with or in the process of obtaining a PhD in virology.	The evaluation suggested that frontier models achieved comparable or better performance to human experts. The best-performing model scored 43.8% whereas PhD-level virologists, assigned VCT question sets specifically tailored to each individual's sub-area of specialization, scored only 22.1%.[42]

39 OpenAI (2024). OpenAI and Los Alamos National Laboratory announce bioscience research partnership. Available: https://openai.com/index/openai-and-los-alamos-national-laboratory-work-together/

40 Anson Ho and Arden Berg (2025), "Do the biorisk evaluations of AI labs actually measure the risk of developing bioweapons?". Published online at epoch.ai. Available: https://epoch.ai/gradient-updates/do-the-biorisk-evaluations-of-ai-labs-actually-measure-the-risk-of-developing-bioweapons

41 Jasper Gotting et al. (2025).Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark. Available: https://arxiv.org/pdf/2504.16137

42 Jasper Gotting et al. (2025).Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark. Available: https://arxiv.org/pdf/2504.16137

Publication	Methodology	Findings/Comment
FutureHouse's "Language Agent Biology Benchmark (LAB-Bench), including ProtocolQA, SeqQA, Cloning Scenarios, and FigQA"	A very broad range of multiple-choice questions evaluating whether or not AI systems can be helpful to biology research assistants, with questions relevant to manipulating biological sequences, designing biological protocols, research-relevant tool use, and interpreting figures	In early 2025, OpenAI's o3 scored 78%, compared with an average score among human experts of 79%. Similarly, in May 2025, Claude Opus 4 performed better than all other models on three of the four evaluations (ProtocolQA, SeqQA, Cloning Scenarios), and reached or exceeded the human baseline in these evaluations.[43]
Oxford Biosecurity Group. "BioLP-bench."[44]	A published benchmark designed to evaluate the ability of language models to find and correct mistakes in a diverse set of laboratory protocols commonly used in biological research. It evaluates model performance on 800 questions from 11 wet lab protocols.	OpenAI's o3-mini and later models exceed the expert baseline performance on this benchmark (38.4%).[45]
"The Weapons of Mass Destruction Proxy (WMDP) benchmark: Measuring and Reducing Malicious Use With Unlearning"[46]	Comprising multiple-choice questions surrounding hazardous knowledge, including biology and chemistry, which serves as a proxy evaluation for hazardous knowledge in foundation models.	Not a specific uplift study per se, rather a public expert-written dataset measuring how LLMs could aid malicious actors in developing biological, cyber, and chemical attack capabilities.

43 Anthropic (2025). System Card: Claude Opus 4 & Claude Sonnet 4. Available: https://www-cdn.anthropic.com/6be99a52cb68eb70eb9572b4cafad13df32ed995.pdf

44 Ivanov, I (2024). BioLP-bench Measuring understanding of AI models of biological lab protocols. Available: https://doi.org/10.1101/2024.08.21.608694

45 OpenAI (2025). Deep Research System Card. Available: https://cdn.openai.com/deep-research-system-card.pdf

46 Li, N et al. (2024) The WMDP Benchmark: Measuring and Reducing Malicious Use With Unlearning. Available: https://arxiv.org/abs/2403.03218

Publication	Methodology	Findings/Comment
OpenAI. "HealthBench Rubric Evaluation."[47]	OpenAI's rubric evaluation, designed to improve health capability measurement AI systems, comprises 5,000 realistic health conversations, each with a custom physician-created rubric to grade model responses.	OpenAI model performance on HealthBech has improved steadily over the last two years (GPT-3.5 Turbo scored 16%, GPT-4o 32% and o3 scored 60%).[48]
SecureBio. "DNA Synthesis Screening Evasion Evaluation."[49]	A test for whether a model can design nucleic acid fragments that assemble into pathogenic viruses by bypassing gene synthesis companies' DNA screening protocols.	In 2025, all of Anthropic's models could design sequences that either successfully assembled plasmids or evaded synthesis screening protocols. None of their models could do both.[50]

Understanding and measuring the capabilities of foundation models is imperative. Similarly, the ability to extrapolate trajectories of development, understand future capabilities, and future-proof risk mitigation strategies is also crucial. Notably, work is underway to build biological general-purpose models that span a wide range of disciplines, including biology, physics, planetary science, and astrophysics, providing the ability to capture, understand, and manipulate the underlying biology contained in complex data. For example, NASA and IBM are building a comprehensive toolkit of fountain models, collectively known as INDUS, which allows scientists to explore new and

47 OpenAI (2025). Introducing HealthBench. Available: https://openai.com/index/healthbench/

48 OpenAI (2025). HealthBench: Evaluating Large Language Models Towards Improved Human Health. Available: https://cdn.openai.com/pdf/bd7a39d5-9e9f-47b3-903c-8b847ca650c7/healthbench_paper.pdf

49 SecureeBio (2025). SecureBio's AI Team: An Overview of Our Biorisk Evaluations. Available: https://securebio.substack.com/p/securebios-ai-team-an-overview-of

50 Anthropic (2025). System Card: Claude Opus 4 & Claude Sonnet 4. Available: https://www-cdn.anthropic.com/6be99a52cb68eb70eb9572b4cafad13df32ed995.pdf

increasingly complex research ideas more efficiently.[51] Building on the developments of their predecessors, these revolutionary capabilities, if realised, will bring with them substantial misuse risks. Safety research to better understand, measure, and extrapolate model capabilities remains essential.

Narrow AI: AI-powered biological tools

Drug discovery is currently undergoing a renaissance fueled by technological and scientific advances that promise to streamline the development of new therapies. The drug discovery landscape is being reshaped by novel computational methods that facilitate the identification and development of new drug candidates.[52] In-silico modeling and simulation techniques are used to predict how drugs will interact with the body, thereby reducing the need for extensive laboratory and animal testing. Additionally, structure-based drug design, empowered by cryo-electron microscopy and advanced computational chemistry, enables the precise design of molecules that can interact more accurately with disease targets.[53] These advancements not only expedite the drug discovery process but also increase its cost-effectiveness, leading to a more efficient path from the laboratory to the patient. This is not a distant future, but a reality that is already shaping the future of the drug discovery pipeline. Not only are AI-designed drugs currently under review or in clinical trials, but some are already advancing towards US

51 NASA (2024). NASA-IBM Collaboration Develops INDUS Large Language Models for Advanced Science Research. Available:https://science.nasa.gov/open-science/ai-language-model-science-research/#:~:text=A%20collaboration%20with%20International%20Business,curated%20scientific%20corpora%20drawn%20from

52 Sliwoski, G et al. (2013). Computational Methods in Drug Discovery. Pharmacological Reviews, December 31, 2013.

53 Zhu, KF (2023). Applications and Prospects of Cryo-EM in Drug Discovery." Military Medical Research 10, Article number: 10. Published March 06, 2023.

FDA approval.[54] The FDA has noted a rapid increase in the number of applications leveraging AI.[55] As major pharmaceutical companies and startups alike further their use of deep learning in their R&D pipelines, the global AI in drug discovery market is expanding rapidly, and the fraction of drugs discovered using AI is expected to increase.[56]

Biological design tools (BDTs) are at the forefront of this renaissance. Biological design tools refer to a range of AI tools trained on biological data using machine learning techniques. In contrast to foundation models, which may be helpful for biological tasks but are not explicitly designed for them, BDTs are specifically developed to assist users with a particular aspect of biological research. These include tools used for designing biological agents, such as proteins or viral vectors, as well as tools for a range of other tasks in life sciences research, models predicting the effects of genetic perturbations, AI-enhanced genome assembly, pathogen property prediction, predictions of protein-protein interactions, and other experimental design or cellular simulation.

The field of highly capable AI-enabled BDTs is growing rapidly across the misuse risk chain. For example, protein design tools may enable the design of novel toxins or those with enhanced stability, while genome assembly tools could enable the assembly of novel viral genomes that may not be flagged by nucleic acid synthesis screening if ordered through synthesis product providers. Further, experimental simulation tools could enable the development of a bioweapon by reducing the

54 InSilico Medicine (2024). Insilico Medicine's AI-designed drug ISM3412 receives FDA IND approval. Available: https://www.news-medical.net/news/20240425/Insilico-Medicines-AI-designed-drug-ISM3412-receives-FDA-IND-approval.aspx

55 U.S. Food and Drug Administration (2025). Artificial Intelligence for Drug Development. Available: https://www.fda.gov/about-fda/center-drug-evaluation-and-research-cder/artificial-intelligence-drug-development

56 Research and Markets (2025). Artificial Intelligence in Drug Discovery Market Outlook Report 2025-2030. Available: https://www.globenewswire.com/news-release/2025/05/08/3076995/28124/en/Artificial-Intelligence-in-Drug-Discovery-Market-Outlook-Report-2025-2030-Accelerating-Drug-Discovery-as-Pharma-Embraces-Machine-Learning-and-Data-Driven-Innovations.html

number of design-build-test-learn cycles required. Notable developments include:

- Outbreak Detection and Forecasting - e.g., EpiWatch, BlueDot
- Variant forecasting tools - e.g., EVEscape
- Protein design tools - e.g., Alphafold 3, AlphaProteo, ProteinBERT
- Small biomolecule design tools - e.g., Chemistry42, MegaSyn
- RNA design -e.g., DeepCRISPR
- In silico testing tools - e.g., Turbine and VCell

While exceptionally beneficial for peaceful pursuits, BDTs could be misused to enable actors across multiple stages in the development of a bioweapon, with the risks increasing rapidly as the pace of technological development accelerates. Governments and industry have initiated efforts to address biological misuse risks posed by foundation models; however, as narrow, specialist AI tools, BDTs will likely require a different set of approaches to mitigate risks.

Additionally, AI BDTs may pose risks that researchers are only beginning to map. For the policy community, the expectation should be that this area will remain one where it is best to anticipate patterns of steady change, along with the potential for sudden and significant leaps forward at any time. For example, as of this writing, much remains unknown regarding how deep learning could uncover genetic vulnerabilities for precision-targeted diseases. Notably, experts have expressed differing opinions on the technical feasibility of precision-targeted treatments for diseases. Today, substantial hurdles persist in actualizing such threats, though many experts have a strong concern that this could change dramatically. As such, this makes it a strong example of how

policymakers must consider different futures that all feature significant and rapid changes in their decision-making.

Applied-AI: the Digital-Physical Frontier

Previously, some experts have argued that oversight of the digital-to-physical divide will be sufficient to reduce many biosecurity risks because the digital information produced by biological models is harmless by itself.[57] However, the rapid developments in AI-enabled biotechnology are casting doubt on this assertion, with many arguing for stronger risk mitigation of this divide.[58] For example, current biological models can generate highly concerning pathogen information or automate substantial portions of the synthesis and rescue process. AI-enabled lab automation and wet lab assistance tools make it easier than ever to bridge the digital-physical divide, particularly when integrated with advances in robotics. Autonomous AI agents optimized for scientific discovery will soon become commonplace in the life sciences, bringing with them vast efficiencies and cost savings, as well as increased biosafety and biosecurity risks.

57 Baker, D & Church, G (2024). Protein design meets biosecurity. Science 383,349-349 DOI:10.1126/science.ado1671

58 Hunter, P (2024). Security challenges by AI-assisted protein design: The ability to design proteins in silico could pose a new threat for biosecurity and biosafety. Science and Society: EMBO Rep 25: 2168 - 2171

DNA Synthesis

The synthetic biology field has witnessed a remarkable rise since the discovery of the Lac Operon, the first gene regulation pattern to be modelled, in 1961.[59] By a 2023 calculation, the international synthetic biology market was expected to surge to $100 billion by 2030.[60] This is also a dynamic field. As of the early months of this study, many analyses concluded that DNA synthesis remained somewhat hindered by relatively slow and error-prone methods, particularly for longer sequences.[61] Less than two years later, emerging AI-enabled technologies and other methodological advances are enhancing DNA synthesis speeds and fidelity, and substantially lowering the costs of doing so.[62]

While this might herald breakthroughs in areas like drug discovery, this progress also brings the risk of enabling the creation of harmful biological agents by teams with less expertise, as evidenced by the 2002 synthesis of the poliovirus even before these tools were available.[63] These could lower the threshold for producing advanced biological agents—a prospect once reserved for well-funded state programs—particularly as benchtop synthesizers become more common. For instance, Benchtop synthesizers are on the cusp of revolutionizing synthetic biology by simplifying the engineering of biological organisms from digital data; however, these technologies bypass traditional biosecurity measures,

59 Cameron, D., Bashor, C. & Collins, J. (2014) A brief history of synthetic biology. Nat Rev Microbiol 12, 381–390. https://doi.org/10.1038/nrmicro3239

60 U.S. Government Accountability Office. "Science & Tech Spotlight: Synthetic Biology." GAO-23-106648, April 17, 2023. https://www.gao.gov/products/gao-23-106648.

61 Daniels, M (2021)."Synthetic DNA in Drug Discovery and Development." Genetic Engineering & Biotechnology News, September 1, 2021.

62 Martin, H et al. (2024). Special Issue on Artificial Intelligence for Synthetic Biology. ACS Synthetic Biology 2024 13 (2), 408-410 DOI: 10.1021/acssynbio.3c00760

63 Name Redacted. "Synthetic Poliovirus: Bioterrorism and Science Policy Implications." CRS Report for Congress, Order Code RS21369, December 13, 2002. Received through the CRS Web.

such as the Select Agent Regulations and the Biological Weapons Convention, which focus on physical sample transfer and containment. Existing frameworks do not adequately cover the synthesis of DNA based on digital information.

Automated Biomanufacturing and Cloud Labs

The biomanufacturing sector is undergoing a transformation fueled by new AI-enabled technologies, including its applications in adaptive manufacturing and cloud laboratories. Adaptive manufacturing integrates traditional production methods with intelligent, flexible systems that can be customized for more personalized and on-demand production. Cloud labs, meanwhile, represent new pathways in research and development, where scientists can access and control laboratory equipment remotely through cloud technology. This digital approach increases efficiency, reduces costs, and accelerates innovation, including by enabling multitasking, facilitating collaborative work among multiple teams from various locations, and optimizing the use of resources and expertise. Simultaneously, this could accelerate investments in distributed manufacturing, which, for the multitude of positive effects, also poses an increased risk in terms of nefarious access to sensitive biological data, processes, and products.[64] Experiments can also be performed remotely through cloud laboratories,[65] further reducing the tacit knowledge requirements for conducting sophisticated research protocols.[66]

64 Nature Biotechnology (2022). Editorial: Increase vigilance against cyberattacks. Nat Biotechnol 40, 1155. https://doi.org/10.1038/s41587-022-01446-4

65 A 2017 review of 1628 scientific papers found 86-89% reported one or more methods that could be conducted in a cloud laboratory – subsequent events that are charaterized in open literature bear out this pattern. Groth, P, and Cox, J (2017). "Indicators for the Use of Robotic Labs in Basic Biomedical Research: A Literature Analysis." PeerJ 5 (November 8, 2017): e3997.

66 World Health Organization (2021). Emerging technologies and dual-use concerns: a horizon scan for global public health. Emerging technologies and dual-use concerns: a horizon scan for global public health.

The integration of big data, cloud computing, and automation has exposed sensitive intellectual property, biological processes, and health and biological data, such as pathogen or patient information, to unprecedented cybersecurity threats. Deloitte's research indicates that in 2019, the life sciences industry was the recipient of the second most frequent targeted cyberattacks across all industries.[67] A particularly alarming risk is the potential leakage of sensitive pathogen information, which could lead to severe biosecurity threats if accessed by malicious entities. This concern is heightened by the incorporation of targeted agentic AI, which could be intentionally targeted at biomanufacturing facilities to disrupt, destroy, extort, or steal sensitive biological research and associated data.

Additionally, the rise in AI-enabled automation, such as multiple tools used in labs, connected instruments, and computer-controlled bioreactors, expands the attack surfaces for malicious actors targeting private or government biological processes.[68] Vulnerabilities in these systems pose a significant risk of operational disruption. Moreover, the high degree of interconnectivity and reliance on third-party integrations within the life sciences supply chain introduces the possibility of sophisticated attacks through these trusted networks. The risks extend beyond mere data breaches to include supply chain attacks, where adversaries exploit vulnerabilities in interconnected systems and networks.

67 Deshpande, R (2023). How to Apply Cybersecurity Best Practices in the Life Sciences Sector," SciLife, May 29, 2023.

68 Olvera, A (2023) The Cyber-Biosecurity Nexus: Key Risks and Recommendations for the United States. Council on Strategic Risks, Briefer No. 54, September 14, 2023.

The Actors

The use of life science technologies for warfare by state and non-state actors has a long history, marked by strategic and tactical evolution. In medieval times, the Spanish and the Mongols intentionally spread disease, while the 20th century saw more structured programs, such as Japan's World War II-era use of plague-infected fleas and the U.S. biological weapons program initiated in 1942.[69] Despite the 1925 Geneva Protocol and the Biological Weapons Convention—the first international treaty to entirely ban biological weapons, submitted for states' ratification in 1972 and entered into force in 1975–state actors have continued to explore biological warfare, showcasing the regulatory challenges in implementing international law. Examples of potential warfare programs include the 1970s-1990s Soviet Union's Biopreparat program, aiming to weaponize pathogens like anthrax, and Iraq's 1980s-1990s bioweapons research. It is widely recognized that some state actors continue to leverage life sciences for military purposes, defying international treaties and norms.[70]

Today, no NATO member country has a biological weapons program. However, given some other states' continued offensive biological activities, Allies have maintained their defensive research on biological agents and continue to invest resources in this area in the context of chemical, biological, radiological, and nuclear defense programs.[71] Unfortunately, for many years, the US Department of Defense has believed "that chemical and biological weapons are likely to be used early in a conflict

69 Frischknecht, F. (2003).. "The history of biological warfare. Human experimentation, modern nightmares and lone madmen in the twentieth century." EMBO Reports, 4(Spec No), S47-S52.

70 U.S. Department of State (2025). Arms Control Compliance Report. Available: https://www.state.gov/wp-content/uploads/2025/04/2025-Arms-Control-Treaty-Compliance-Report-1.pdf

71 NATO (2023). Weapons of mass destruction. Available: https://www.nato.int/cps/en/natohq/topics_50325.htm

to disrupt US operations and logistics and to potentially offset the overwhelming conventional warfare capabilities of US forces."[72]

Current biotechnological advancements broaden the biological warfare arsenal beyond natural pathogens, with genetic engineering tools like CRISPR/Cas9 enabling the creation of more potent biological agents. [73] Advances in technology could also allow for a more targeted delivery of biological weapons through increasingly sophisticated delivery mechanisms such as tailored drones or nano-robots. Advances in peptide synthesis could also enable sophisticated actors to develop a wide range of novel toxins with both incapacitating and lethal effects that are not on a select agent list.[74]

In parallel to these developments, the West has witnessed a stark increase in gray zone malicious activity. Hostile actors are now able to obfuscate bioweapons behind complex molecular smokescreens through tactics such as, among other things, mimicking naturally occurring mutations that bear no signatures of synthetic creation. Disturbingly, some countries are actively concealing what may be bioweapons research in plain sight, intertwining it with their growing biotechnology industries and civilian research. Future biological warfare in gray zones could see nations seeking to bypass traditional bioweapons constraints, including by attempting small-scale outbreaks in smaller, less conspicuous settings like industrial or academic labs, or

72 U.S. Government Accountability Office (2000). Chemical and Biological Defense: Units Better Equipped, but Training and Readiness Reporting Problems Remain." GAO-01-27. Published: Nov 14, 2000. Available: https://www.gao.gov/assets/gao-01-27.pdf.

73 U.S. Department of State (2025). Arms Control Compliance Report. Available: https://www.state.gov/wp-content/uploads/2025/04/2025-Arms-Control-Treaty-Compliance-Report-1.pdf

74 U.S. Department of Defense (2023) Biodefense Posture Review. Available: https://media.defense.gov/2023/Aug/17/2003282337/-1/-1/1/2023_biodefense_posture_review.pdf

around logistics hubs.[75] Modern bioweapons may therefore be more subtle, and not necessarily human-targeted, instead used to deny, degrade, and disrupt our agriculture, trade, and natural habitat, in addition to the potential for larger-scale and human-targeted threats.

Competitors and Adversaries

AI is fast becoming a new axis of state competition. Indeed, America's AI Action Plan, published in July 2025, commits to "winning the AI race" to protect the US's economic and military leadership.[76] To comprehend this emerging landscape, one must examine the competitors in the race. The following section sets out the US's key competitors and adversaries in the global AI race.

China

China is employing an aggressive, whole-of-government approach, combined with state guidance of the private sector, to become a global S&T superpower, prioritizing technology sectors such as advanced energy, AI, biotechnology, quantum information science, semiconductors, and advanced technologies for military modernization. To leapfrog over technological hurdles, China has systematically stolen thousands of gigabytes of intellectual property from companies in Asia, Europe, and North America.[77] Even setting aside any potential intentions to weaponize these assets, this presents a range of strategic advantage questions to other leading nations.

75 Gisselsson D. (2022). Next-Generation Biowarfare: Small in Scale, Sensational in Nature? Health Secur. Mar-Apr;20(2):182-186. doi: 10.1089/hs.2021.0165. Epub 2022 Jan 12. PMID: 35021880; PMCID: PMC9080995.

76 Winning the Race: America's AI Action Plan, (2025), https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf.

77 U.S. Office of the Director of Intelligence (2025). Annual Threat Assessment of the U.S. Intelligence Community. Available: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2025-Unclassified-Report.pdf

The US government assesses publicly that the PRC military medical institutions conducted toxin and biotechnology research and development with potential biological weapons applications.[78] The PRC's 14th Five-Year Plan for Bioeconomy Development identifies DNA sequencing, gene editing, and synthetic biology as examples of cutting-edge biotechnology. The People's Liberation Army (PLA)'s medical institutions have established themselves as "major centers for research in gene editing and other new frontiers of biotechnology."[79] The PRC's China Brain Project (2016–2030) aims to understand cognitive functions and neural pathways, focusing on R&D of various dual-use technologies, including brain-computer interfaces (BCI).[80] PLA scientists have investigated various military applications of brain research, including enhanced mental agility, improved mobility, physical safety, and situational awareness for combatants. The PLA is exploring a range of "neurocognitive warfare" capabilities that exploit adversaries using neuroscience and psychology.[81]

The PRC's dominance in pharmaceutical supply production, thanks in part to lower quality, safety, and environmental standards, has positioned Beijing to restrict such exports for leverage over Washington and

78 U.S. Department of State (2024). Adherence To And Compliance With Arms Control, Nonproliferation, And Disarmament Agreements And Commitments. Available: https://2021-2025.state.gov/wp-content/uploads/2024/04/2024-Arms-Control-Treaty-Compliance-Report.pdf

79 Keown, A (2020). Is CRISPR in use to Create China Super Soldiers? Available: https://www.biospace.com/is-china-using-crispr-to-create-super-soldiers

80 Brain-computer interfaces (BCI) are an emerging technology which enables humans and computers to exchange information through devices implanted in the brain or placed on the skull.

81 U.S. Department of Defense (2024). Military And Security Developments Involving The People's Republic Of China Annual Report To Congress. Available: https://media.defense.gov/2024/Dec/18/2003615520/-1/-1/0/MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2024.PDF

others in trade or security disputes.[82, 83] Beijing has explicitly identified genetic data as a "national strategic resource" and is expanding state control over the country's genetic repositories, positioning it to potentially lead in precision medicine and agricultural biotechnology applications.[84]

As noted in the US Department of Defense's 2024 report to Congress:[85]

> "The PRC is prioritizing the development of AI-enabled capabilities because of its belief that AI is leading to the next revolution in military affairs. To actualize the level of AI integration that the PLA envisions, Beijing recognizes the need to leverage developments from across its commercial and academic sectors. By 2030, the PLA expects to field a range of "algorithmic warfare" and "network-centric warfare" capabilities operating at different levels of human-machine integration. CCP leaders believe AI and machine learning will enhance information, surveillance, and reconnaissance capabilities, enabling a range of new defense applications, including autonomous and precision-strike weapons."

The PLA aspires to integrate AI into its future C4ISR (Command, Control, Communications, Computers & Intelligence, Surveillance and Reconnaissance) capabilities, processes, and systems, including through reconnaissance data processing, automated target recognition,

82 U.S. Office of the Director of Intelligence (2025). Annual Threat Assessment of the U.S. Intelligence Community. Available: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2025-Unclassified-Report.pdf

83 Suarez, V (2024). The National Security Rationale for Stockpiling Key Pharmaceutical Ingredients. Available: https://councilonstrategicrisks.org/2024/03/05/the-national-security-rationale-for-stockpiling-key-pharmaceutical-ingredients/

84 U.S. Office of the Director of Intelligence (2025). Annual Threat Assessment of the U.S. Intelligence Community. Available: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2025-Unclassified-Report.pdf

85 U.S. Department of Defense (2024). Military And Security Developments Involving The People's Republic Of China. Available: https://media.defense.gov/2024/Dec/18/2003615520/-1/-1/0/MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2024.PDF

and command & control decision-making support. The PRC is pursuing domestic production of critical technologies to reduce reliance on foreign sources.[86]

Russia

Russia maintains an offensive biological weapons (BW) program. Russia's 48th Central Scientific Research Institute facilities, considered a key asset within its offensive biological weapons activities, have recently been modernized, raising serious concerns about Moscow's broader programmatic ambitions.[87] Following its 2022 invasion of Ukraine, Russia's science and technology ecosystem has faced significant constraints, primarily due to international sanctions that limit access to Western technology. Despite these impediments, the Russian state continues to operationalize emerging AI applications in both military and civilian contexts. Concurrently, Russia has strengthened its technical cooperation with international partners, notably China, to advance its long-term research and development objectives.

A key application of Russian AI is the generation of sophisticated synthetic media (deepfakes) for disinformation campaigns and malign influence operations. In the domestic microelectronics sector, Russia's manufacturing capabilities are currently limited to the 65nm semiconductor process node, with a stated goal of achieving 28nm mass production by 2030. This timeline represents a substantial technological deficit compared to global industry leaders. To mitigate its isolation from Western supply chains, Russia is strategically aligning its S&T

86 U.S. Department of Defense (2024). Military And Security Developments Involving The People's Republic Of China Annual Report To Congress. Available: https://media.defense.gov/2024/Dec/18/2003615520/-1/-1/0/MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2024.PDF

87 U.S. Department of State (2024). Adherence To And Compliance With Arms Control, Nonproliferation, And Disarmament Agreements And Commitments. Available: https://2021-2025.state.gov/wp-content/uploads/2024/04/2024-Arms-Control-Treaty-Compliance-Report.pdf

initiatives with China and other BRICS nations (Brazil, India, South Africa). This collaboration focuses on areas such as AI development, technology governance, and semiconductor production, with the dual aims of enhancing indigenous capabilities and reducing the global technological influence of Western countries.[88]

The DPRK

The DPRK has a dedicated, national, offensive BW program, capable of producing biological agents such as bacteria, viruses, and toxins for military purposes. As reported by the US State Department, "The DPRK also has the capability to genetically engineer biological products with technologies such as CRISPR, as reported by its State Academy of Sciences and other sources. Pyongyang is probably capable of weaponizing BW agents with unconventional systems such as sprayers and poison pen injection devices, which have been deployed by the DPRK for the delivery of chemical weapons and could be used to covertly deliver BW agents."[89] The DPRK continues to improve its capabilities by collaborating with other countries on dual-use scientific topics or procuring biological equipment and materials.[90]

88 U.S. Office of the Director of Intelligence (2025). Annual Threat Assessment of the U.S. Intelligence Community. Available: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2025-Unclassified-Report.pdf

89 U.S. Department of State (2024). Adherence To And Compliance With Arms Control, Nonproliferation, And Disarmament Agreements And Commitments. Available: https://2021-2025.state.gov/wp-content/uploads/2024/04/2024-Arms-Control-Treaty-Compliance-Report.pdf

90 U.S. Department of State (2024). Adherence To And Compliance With Arms Control, Nonproliferation, And Disarmament Agreements And Commitments. Available: https://2021-2025.state.gov/wp-content/uploads/2024/04/2024-Arms-Control-Treaty-Compliance-Report.pdf

Iran

Iran has historically engaged in dual-use activities with potential military applications. Open-source reporting suggests that Iranian military-associated academic institutions continue to pursue BW-relevant projects on bioregulators.[91]

Non-state Actors

Non-state actors have shown interest in acquiring biological manipulation capabilities for bioterrorism activities over the past decades. Key incidents exemplify this trend, each with unique motivations and methodologies. In 1984, the pseudo-religious cult known as the Rajneeshees tried to rig a local election by incapacitating voters with deliberately contaminated food at salad bars and other eateries at The Dalles, Oregon. The Japanese pseudo-religious cult known as Aum Shinrikyo sought to save the world through destruction by producing and releasing chemical and biological weapons in Japan in the 1980s and 1990s.

In 2001, letters containing anthrax spores were posted to several US news media offices and US Senators, resulting in 22 cases of anthrax infection, with five fatalities.[92] Since 2001, numerous non-state actors have continued to show a keen interest in researching, developing, scaling, and deploying biological weapons. A list of pathogens was found in 2002 by US Navy Seals in an Afghanistan cave that Al Qaeda intended to use in a series of biological attacks, including those on human, animal, and agricultural targets. These concerns are exacerbated as state-sponsored terrorist organizations have emerged in recent years: a combination that provides the high-tech capabilities

91 U.S. Department of State (2024). Adherence To And Compliance With Arms Control, Nonproliferation, And Disarmament Agreements And Commitments. Available: https://2021-2025.state.gov/wp-content/uploads/2024/04/2024-Arms-Control-Treaty-Compliance-Report.pdf

92 NPR Staff (2011) Timeline: How The Anthrax Terror Unfolded." NPR, February 15, 2011.

and resources of nation-states mixed with the anonymity and zeal of a terrorist organization.

Modern biological weapons facilities, whether from state or non-state actors, may not have the same characteristics, objectives, or goals as past bioweapons programs have. Combining emerging technologies and progress in areas such as automation, additive manufacturing, and cloud labs, among others, likely means that modern biological weapons programs from non-state actors are likely to be small and targeted towards specific purposes. Although bioweapons are classified under the category of weapons of mass destruction, more recent examples of non-state actors' BW use display a tendency to use them more to elicit panic and have governments waste resources, or for small-scale and assassination purposes.

Risk Mitigation Responses

The scientific community is divided on the extent of risk associated with AI-enabled biotechnology, resulting in a myriad of policy responses. These differences often stem from disparate views on the interpretation and prioritization of risks, divergent approaches to science policy, and the weighting of theoretical evidence versus empirical data. Factors such as varying risk tolerance levels, differing ethical perspectives on genetic modification, and contrasting beliefs about the role of precaution in scientific advancement also significantly contribute to these disagreements. There can also be significant variance based on the tools different experts use and how they use them, given the growing variety of AI-enabled capabilities available. Such diversity in viewpoints highlights the complexity and multidimensional nature of assessing and managing risks associated with AI-enabled biotechnology.

Multilateral Initiatives

- **The Bletchley AI Safety Summit 2024, the AI Seoul Summit 2024, and the AI Action Summit Paris, February 2025.** The first AI Safety Summit initiated the Bletchley Declaration, the State of the Science Report (now titled the International Scientific Report on Advanced AI Safety), and led to a commitment to pre-deployment testing of AI models, ultimately resulting in the establishment of AI Safety Institutes worldwide. The Bletchley Declaration set out to:
    - » Identify common AI risks while building a shared and sustained scientific understanding of such risks, grounded in evidence.

- » Develop risk-based policies across a global community, collaborating as appropriate while acknowledging differences in national and regional approaches.
- » Increase transparency from private actors developing frontier AI capabilities.
- » Establish relevant public sector capabilities and scientific research capabilities to collaborate with industry on the development of appropriate evaluation metrics and tools for safety testing.
- » Sustain an inclusive international dialogue, engaging existing international fora and other relevant initiatives.

- At the AI Seoul Summit 2024, leading AI organisations agreed to the voluntary Frontier AI Safety Commitments in furtherance of safe and trustworthy AI. The Commitments included effectively identifying, assessing and managing risks when developing and deploying frontier AI models and systems, as well as accountability and transparency commitments.[93]
- The February 2025 AI Action Summit in Paris marked a shift from a safety focus to an emphasis on opportunity, with France announcing that it had secured commitments to invest more than €109 billion over the next few years.[94] The final declaration was mostly signed by European countries, with the UK and the US noticeably absent, which has been attributed to a mix of insufficient governmental coordination, a lack of practical clarity on global governance, and missing the harder national security questions.

93 Frontier AI Safety Commitments, AI Seoul Summit 2024, Gov. UK. Available: https://www.gov.uk/government/publications/frontier-ai-safety-commitments-ai-seoul-summit-2024/.

94 Piquard, A (2025). "AI: With announcement of investments worth €109 billion, Macron intends to take on US," February 10, 2025, Le Monde. Available: https://www.lemonde.fr/en/economy/article/2025/02/10/ai-with-the-announcement-of-a-109-billion-investment-macron-intends-to-take-on-the-us_6737985_19.html.

- India will host the next "AI Impact Summit" in New Delhi in February 2026.

Industry Initiatives

- **Frontier Model Forum (FMF).** An industry-supported non-profit dedicated to advancing frontier AI safety and security, launched by Anthropic, Google, Microsoft, and OpenAI, and now also including Amazon and Meta. The FMF aims to identify best practices and support standards development, advance science and independent research, and facilitate information sharing among government, academia, civil society, and industry.[95] As an avenue for exchange on frontier capability assessments, the Forum has gathered these key players' current practices for determining whether frontier models have capabilities that could increase risks to public safety and security, such as by facilitating the development of chemical, biological, radiological, or nuclear (CBRN) weapons.[96] In April 2025, the Forum kicked off a series of technical reports on frontier AI safety frameworks to provide detailed insight into key components of these frameworks, incorporating lessons from early adopters while acknowledging areas where best practices continue to evolve.[97]
- **Voluntary deployment frameworks and safety policy.** Safety policy frameworks are crucial for ensuring the responsible development and deployment of AI systems.

95 Frontier Model Forum (2024). About. Available: https://www.frontiermodelforum.org/about-us/

96 Frontier Model Forum (2024). Frontier-capability-assessments. Available: https://www.frontiermodelforum.org/technical-reports/frontier-capability-assessments/

97 Frontier Model Forum (2025). Technical Report Frontier-capability-assessments. Available: https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf

These frameworks address various concerns, including potential biases, harmful outputs, data privacy, and security risks associated with LLMs. A leading example of this is Anthropic's Responsible Scaling Policy, which provides a framework for managing the risks associated with increasingly powerful foundation models. It outlines technical and organizational protocols to mitigate potential harms, focusing on catastrophic risks such as the misuse of technology for the development of dangerous weapons and autonomous actions contrary to design. Similarly, OpenAI's Preparedness Framework is a process for "tracking and preparing for advanced AI capabilities that could introduce new risks of severe harm."[98] Other leading AI developers have published risk management frameworks such as xAI's Risk Management Framework[99] and DeepMind's Frontier Safety Framework.[100]

- **Commissioning Deep Safety Research and Benchmarking.** Large AI labs and a handful of smaller developers are actively investing in and commissioning more exhaustive AI safety research to establish best practices for the responsible development and deployment of foundation models. They are also cultivating partnerships with policy-makers and academia to communicate insights regarding trust and safety risks effectively. AI benchmarking is crucial for objectively evaluating and comparing different AI systems, as well as assessing their capabilities in relation to countering CBRN misuse. For example, the Virology Capabilities Test is an LLM benchmark that measures capabilities related to

98 OpenAI (2025). Preparedness Framework. Available: https://cdn.openai.com/pdf/18a02b5d-6b67-4cec-ab64-68cdfbddebcd/preparedness-framework-v2.pdf

99 https://x.ai/documents/2025.02.20-RMF-Draft.pdf

100 DeepMind (2024). Frontier Safety Framework. Available: https://storage.googleapis.com/deepmind-media/DeepMind.com/Blog/introducing-the-frontier-safety-framework/fsf-technical-report.pdf

troubleshooting complex virology laboratory protocols; it remains a crucial predeployment risk mitigation tool.[101]

- **Capability Evaluation and Redteaming.** External evaluations of powerful foundation models remain essential, and national AI Safety Institutes should be effectively resourced to expand this vital impartial capability. For example, OpenAI and Los Alamos National Laboratory are co-developing novel evaluations to enhance understanding of how multimodal AI models can be used responsibly in laboratory settings.[102] AI redteaming is crucial for identifying vulnerabilities in AI systems, especially those related to CBRN threats. By simulating adversarial attacks and stress-testing AI models, red teaming helps uncover weaknesses and potential risks that malicious actors could exploit, strengthening the system's defenses and promoting safety. The large AI labs have developed internal red teams. For example, Anthropic's Frontier Red Team was established to explore whether AI models have developed critical, advanced capabilities and to help mitigate risks associated with them before deployment.
- **Early Warning Systems and Security Protocols.** AI companies are exploring the development of early warning systems built into their protocols, designed as a tripwire to alert security officers to the potential existence of CBRN weapons development capabilities. Furthermore, OpenAI has established an insider threat protection function to better protect its intellectual property and mitigate the risk

101 Jasper Gotting et al. (2025).Virology Capabilities Test (VCT): A Multimodal Virology Q&A Benchmark. Available: https://arxiv.org/pdf/2504.16137

102 OpenAI (2024). OpenAI and Los Alamos National Laboratory announce bioscience research partnership. Available: https://openai.com/index/openai-and-los-alamos-national-laboratory-work-together/

of capability theft and optimization for nefarious means.[103] Anthropic has recognised the potential insider threats posed by agentic misalignment and is monitoring this potential threat as the models advance.[104]

- **Securing the Physical/Digital Frontier.** AI-enabled technologies are rapidly reducing the barriers to entry for translating in-silico design to in-vitro development, whether through lab automation or the ability to 'print' nucleic acids with virtually any sequence. Security researchers, AI labs, and policy makers are increasingly aware of the need to better understand and regulate this critical frontier. DNA Synthesis Screening[105] is recognized as an essential mechanism for preventing or mitigating the accidental or deliberate misuse of engineering biology.[106] Some AI labs are incorporating evaluations to understand their models' capabilities in this regard. For example, Anthropic has incorporated SecureBio's "DNA Synthesis Screening Evasion" evaluation tests into their predeployment evaluations.[107]

103 OpenAI (2025). Careers: Security Engineer, Insider Threat Detection and Response. Available https://openai.com/careers/security-engineer-insider-threat-detection-and-response/

104 Anthropic (2025). Agentic Misalignment: How LLMs could be insider threats. Available: https://www.anthropic.com/research/agentic-misalignment

105 Tessa Alexanian & Sella Nevo. "Supporting Follow-Up Screening for Flagged Nucleic Acid Synthesis Orders." May 7, 2024. Briefer, Nolan Center. Council on Strategic Risks. Available at https://councilonstrategicrisks.org/2024/05/07/supporting-follow-up-screening-for-flagged-nucleic-acid-synthesis-orders/.

106 Guidance: UK screening guidance on synthetic nucleic acids for users and providers, October 8, 2024, https://www.gov.uk/government/publications/uk-screening-guidance-on-synthetic-nucleic-acids/uk-screening-guidance-on-synthetic-nucleic-acids-for-users-and-providers.

107 Anthropic, System Card: Claude Opus 4 & Claude Sonnet 4, (2025), https://www-cdn.anthropic.com/4263b940cabb546aa0e3283f35b686f4f3b2ff47.pdf.

U.S. Government Actions

Policy attention to the AIxBio nexus has grown significantly over the past two years, coinciding with the general rise in attention to AI policy, as evidenced by several relevant executive branch actions. While there have been significant shifts in terms of larger AI policy, the AIxBio policy landscape remains more consistent than others, given a bipartisan national security interest. Still, specific AIxBio policy actions are constantly evolving, reflecting the dramatic developments in AI capabilities. This policy environment has also seen crucial public-private cooperation that will shape the direction of policy in the years ahead. Some of these interactions and shifts during the past few years are traced in the sections below.

The Recent Past

AIxBio has been an extremely active policy space over the past few years, with numerous concepts and measures advancing into executive orders and agency guidance, while proposed legislation has primarily remained in draft form. In hindsight, one catalytic measure was the federal government securing voluntary commitments from leading AI companies that included "internal and external red-teaming of models or systems in areas including misuse, societal risks, and national security concerns, such as bio."[108] As of this writing, these major companies have been actively meeting this pledge by hiring internal experts to conduct risk analysis and mitigation, convening outside experts to red-team models before release, and facilitating government-industry collaboration on AI model evaluations through the U.S. AI Safety Institute.

108 The White House (2023). Ensuring Safe, Secure, and Trustworthy AI. Available: https://bidenwhitehouse.archives.gov/wp-content/uploads/2023/07/Ensuring-Safe-Secure-and-Trustworthy-AI.pdf

Current Policy Context

As of this writing, the US approach to AI safety has undergone a significant transformation within a matter of months. For AI in general, this is marked by a decisive pivot from a cautionary, regulation-focused stance to a strategy centered on accelerating innovation and ensuring American dominance in the global AI race. During his first week in office, President Trump issued an Executive Order (EO) revoking the Biden administration's 2023 EO 14110 on "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence," and directing his top AI officials to develop a comprehensive AI Action Plan for the federal government. The Action Plan was to achieve the EO-established "policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security."[109]

This pivot fully materialized in July 2025 with the introduction of the "Winning the AI Race: America's AI Action Plan" by the current administration. The new policy framework prioritizes the rapid development and deployment of AI technologies by removing what the administration deems "burdensome" regulations. The core tenets of the action plan revolve around three pillars: accelerating innovation, building robust AI infrastructure, and leading in international AI diplomacy and security. This represents a move away from the previous administration's emphasis on the ethical implications of AI, including concerns about bias, misinformation, and potential societal harms.

The AI Action Plan includes a chapter on investing in biosecurity, which recognizes that AI could create new pathways for malicious actors to synthesize harmful pathogens and other biomolecules—and

109 The White House (2025). Removing Barriers To American Leadership In Artificial Intelligence. Available: https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence/

views as the solution to this problem a multitiered approach designed to screen for malicious actors, along with new tools and infrastructure for more effective screening, and notes the need to work with allies and partners for international adoption. It includes provisions requiring all institutions receiving Federal funding for scientific research to use nucleic acid synthesis tools and synthesis providers that have robust nucleic acid sequence screening and customer verification procedures—including enforcement mechanisms. Moreover, it asks OSTP to convene government and industry actors to develop a mechanism to facilitate data sharing between nucleic acid synthesis providers to screen for potentially fraudulent or malicious customers. Last but not least, it includes a recommended policy action to build, maintain, and update as necessary national security-related AI evaluations through collaboration between CAISI at DOC, national security agencies, and relevant research institutions.[110]

On the specific nexus of biological risks and AI, it marks an important policy indicator that the Trump administration's AI Action Plan is clear about the potential threats and instructs multiple, specific actions to help address those risks. This prioritization extends to specific AIxbio actions, which will be led by the Center for AI Standards and Innovation. A Biden-era National Security Memorandum (NSM) initially tasked the AI Safety Institute (AISI) within the National Institute of Standards and Technology (NIST) under Commerce to "serve as the primary United States Government point of contact with private sector AI developers to facilitate voluntary pre- and post-public deployment testing for safety, security, and trustworthiness of frontier AI models."[111] It tasked Commerce, in coordination with relevant

110 The White House (2025). Winning the Race: America's AI Action Plan. Available: https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf.

111 Sharre, P et al. (2024). Noteworthy: "Memorandum on Advancing the United States' Leadership in Artificial Intelligence". Available: https://www.cnas.org/press/noteworthy/noteworthy-memorandum-on-advancing-the-united-states-leadership-in-artificial-intelligence?x-craft-preview=lfcfowvfek

agencies as appropriate, to "establish an enduring capability to lead voluntary unclassified pre-deployment safety testing of frontier AI models on behalf of the United States Government, including assessments of risks relating to cybersecurity, biosecurity, chemical weapons, system autonomy, and other risks as appropriate...".[112] It extended to classified evaluations of the advanced AI models' capacity to generate or exacerbate deliberate biological threats, to which DHS and DOE primarily contributed expertise, information, and infrastructure.

On the other hand, the NSM also recognized opportunities to leverage AI for bio-risk mitigation by tasking agencies pursuing the development of AI systems substantially trained on biological and chemical data to, as appropriate, support efforts to utilize high-performance computing resources and AI systems to enhance biosafety and biosecurity. These efforts included the development of tools for in silico screening of chemical and biological research and technology; the creation of algorithms for nucleic acid synthesis screening; the construction of high-assurance software foundations for novel biotechnologies; the screening of complete orders or data streams from cloud labs and biofoundries; and the development of risk mitigation strategies such as medical countermeasures.

Among other areas, the Department of Defense began working toward the goal of integrating AI into plans to address biological risks. The GUIDE program, for example, is an interagency program that aims "to leverage its integrated computational and experimental capabilities to accelerate drug development for the warfighter by harnessing the power of advanced simulation and machine learning."[113] The Department is

112 Sharre, P et al. (2024). Noteworthy: "Memorandum on Advancing the United States' Leadership in Artificial Intelligence". Available: https://www.cnas.org/press/noteworthy/noteworthy-memorandum-on-advancing-the-united-states-leadership-in-artificial-intelligence?x-craft-preview=lfcfowvfek

113 U.S. Department of Defense (2025). GUIDE Program Prepares DOD to Defend Against the Unknown. Available: https://www.army.mil/article/274069/guide_program_prepares_dod_to_defend_against_the_unknown

also collaborating with international partners to integrate interoperability into AI tools and capabilities. For example, the Japanese Ministry of Defense's Acquisition, Technology & Logistics Agency (ATLA) and the US Department of Defense's Defense Innovation Unit (DIU) recently hosted the US-Japan Global Innovation Challenge 2025, with the aim of "advancing biodefense solutions and AI-driven command and control and decision making."[114]

Under the direction of President Trump, Secretary of Commerce Howard Lutnick announced earlier this year his plans to reform the agency formerly known as the US AI Safety Institute (AISI) into the Center for AI Standards and Innovation (CAISI). CAISI is still tasked with establishing voluntary agreements with private sector AI developers and evaluators, and leading unclassified evaluations of AI capabilities that may pose risks to national security. The administration's guidance includes focusing on "demonstrable risks, such as cybersecurity, biosecurity, and chemical weapons." The AI Action Plan also reinforces CAISI's role in evaluating frontier AI systems for national security risks in partnership with frontier AI developers and other agencies with relevant expertise in CBRNE and cyber risks. It provides direction for prioritizing the recruitment of leading AI researchers at Federal agencies, including NIST and CAISI within DOC, DOE, DOD, and the IC, to ensure that the Federal government can continue to offer cutting-edge evaluations and analysis of AI systems.

To coordinate CAISI's efforts internationally, the International Network of AI Safety Institutes launched in November 2024, comprising the "AI safety institutes and government mandated offices that facilitate AI safety and evaluation from Australia, Canada, the European Commission, France, Japan, Kenya, the Republic of Korea,

114 U.S. Department of Defense (2025).U.S.-Japan Global Innovation Challenge https://nsin.mil/events/2025-01-22-us-japan-joint-challenge/

Singapore, the United Kingdom, and the United States."[115] Affirming and building on the Seoul Statement of Intent toward International Cooperation on AI Safety Science released at the AI Seoul Summit on May 21, 2024, the network aims "to catalyze a new phase of international cooperation on AI safety." [116]

International Responses

Governments have responded differently to the potential threats posed by AI, resulting in a diverse and fragmented policy landscape. Encouragingly, several governments have already adopted policies that require them to assess the potential risks and impacts of new AI models and tools, and have established risk thresholds where further mitigation or response is necessary. The following section provides a brief description of the EU, the UK, China, and NATO's responses, highlighting a few key mitigation measures that are shaping the AIxbio landscape.

The European Union

The European Union has taken a pioneering and comprehensive approach to AI safety, centered around the EU AI Act, which entered into force in August 2024. The EU's strategy is based on a risk-based approach that aims to ensure AI systems are safe, ethical, and trustworthy, while promoting innovation and responsible development. The EU's risk mitigation responses include:

115 International Network of AI Safety Institutes, Mission Statement, November 2024, available at https://www.nist.gov/system/files/documents/2024/11/20/Mission%20Statement%20-%20International%20Network%20of%20AISIs.pdf.

116 International Network of AI Safety Institutes, Mission Statement, November 2024, available at https://www.nist.gov/system/files/documents/2024/11/20/Mission%20Statement%20-%20International%20Network%20of%20AISIs.pdf.

- The 2024 EU AI Act applies to businesses operating in the EU and, critically, the tech giants behind the AI products used in the EU. The risk-based approach requires tighter regulatory scrutiny of AI in ascending order of the potential risks involved in their application.
- The European AI Office was established to monitor compliance with the AI Act.[117] Its mandate includes assessment of the risks posed by foundation models and explicitly includes CBRN risks as one of the significant systemic risks.[118]
- The GenAI4EU was launched in 2024 to "support startups and SMEs in developing trustworthy AI that complies with EU values and rules." Biotech is a key Application area for the initiative.[119]
- In March 2025, the Commission launched a public consultation on the forthcoming EU Bioeconomy Strategy, which is expected to be adopted by the end of 2025. The strategy aims to "advance innovation and maintain the EU's leadership in the bioeconomy. It will propose actions to unlock the potential of bioeconomy innovations, so that they can reach the market, generating green jobs and growth."[120]

Taking effect on August 2, 2025, pending formal approval by the AI Board during its adequacy assessment, the European Union's newly finalized Code of Practice for general-purpose AI models has transformed the AI Act's high-level requirements into concrete

117 European Commission (2024). European AI Office. Available: https://digital-strategy.ec.europa.eu/en/policies/ai-office

118 European Commission (2025). EU AI Office Launches €9 million tender for Technical Support on GPAI Safety. Available: https://digital-strategy.ec.europa.eu/en/funding/eu-ai-office-launches-eu9-million-tender-technical-support-gpai-safety

119 European Commission (2024). European AI Office. Available: https://digital-strategy.ec.europa.eu/en/policies/ai-office

120 European Commission (2025). EU Biotech Act | Dual-Use and Security Concerns. Available: https://www.eu-biotech-act.com/

standards to support industry in preparing for the AI Act's rules on general-purpose AI. A voluntary framework, the Code was developed by 13 independent experts with input from over 1,000 stakeholders, including AI providers, SMEs, academics, civil society, and AI safety experts. Signatories are expected to understand systemic risks associated with some GPAI models, estimate timelines for when models might exceed the highest risk tiers, and take steps to monitor and mitigate risks. A few key industry players like OpenAI and Mistral have already announced their intention to sign onto the Code.[121] The voluntary standards will be enforced within a year for new models, while existing models will have two years to comply with the rules.[122]

The UK

The potential risks from the convergence of AI and biotechnology have been widely discussed and were recognized early by the UK Government. However, the current government has initiated a shift in the UK's AI strategy, moving away from global leadership on AI safety initiatives towards domestically oriented, growth-focused initiatives that focus on public sector adoption of AI. Despite this, the UK retains a world-leading capability in AISI, which continues to provide partner governments with a scientific analysis of the risks posed by advanced AI.

The UK's risk mitigation responses include:

- The world's first AI Safety Institute was established in September 2023. The UK AI Safety Institute (now the AI Security Institute) focuses on advanced AI safety for the public interest. AISI is currently evaluating the capabilities

121 OpenAI (2025). The EU Code of Practice and future of AI in Europe. Available: https://openai.com/global-affairs/eu-code-of-practice/

122 European Commission (2025) The General-Purpose AI Code of Practice. Available: https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai#:~:text=EU%20copyright%20law.-,Safety%20and%20Security,-The%20Safety%20and

most relevant to enabling malicious actors, such as aiding in cyber-criminality, biological or chemical science, human persuasion, large-scale disinformation campaigns, and weapons acquisition.[123] AISI continues to pioneer research on the security implications of AI and, in February 2025, announced it was working with a wide range of national security stakeholders to assess the national security risks posed by frontier AI.[124]

- In November 2023, the UK hosted the inaugural AI Safety Summit, The Bletchley Summit, which positioned the UK at the forefront of AI safety discussions and regulation.
- Diverging with the EU, the UK decided against blanket legislation; instead, it adopted a non-statutory, contextual, cross-sectoral principles-based approach.[125] These principles were set out in a 2023 White Paper[126] as: "Safety, security, and robustness; appropriate transparency and explainability; fairness; accountability and governance; and contestability and redress."[127]

123 AI Security Institute (2025) AISI: About. Available: https://www.aisi.gov.uk/about

124 UK Department for Science, Innovation and Technology (2025) Tackling AI security risks to unleash growth and deliver Plan for Change. Available: https://www.gov.uk/government/news/tackling-ai-security-risks-to-unleash-growth-and-deliver-plan-for-change

125 UK Department for Science, Innovation and Technology (2024). Consultation Outcome: A pro-innovation approach to AI regulation: government response. Available: https://www.gov.uk/government/consultations/ai-regulation-a-pro-innovation-approach-policy-proposals/outcome/a-pro-innovation-approach-to-ai-regulation-government-response

126 UK Department for Science, Innovation and Technology (2024). Consultation Outcome: A pro-innovation approach to AI regulation: government response. Available: https://www.gov.uk/government/consultations/ai-regulation-a-pro-innovation-approach-policy-proposals/outcome/a-pro-innovation-approach-to-ai-regulation-government-response

127 DAC Beachcroft (2025). The approach to the regulation of AI in the UK. Available: https://www.dacbeachcroft.com/The-approach-to-the-regulation-of-AI-in-the-UK

- Instead of a centralized regulator, the current UK framework devolves authority and responsibility to existing regulators, leveraging their expertise to oversee AI within their respective sectors. The Department for Science, Innovation, and Technology (DSIT) published new guidance for regulators to support them when interpreting and applying the principles-based approach.
- In November 2024, the UK launched the Laboratory for AI Security Research (LASR) tasked to "mitigate risks to and from AI, drive national resilience in AI skills and position the UK as a global leader in AI security and governance" through cutting-edge research.[128]

NATO

Though not involved in civil regulation, the Alliance is increasingly active in addressing AI safety and responsible AI development, as it has recognized both the immense opportunities and potential risks of AI for defense and security. The inaugural 2021 AI Strategy set the Alliance's strategic vision for AI, emphasizing six "Principles of Responsible Use of AI in Defense: Lawfulness, Responsibility and Accountability, Explainability and Traceability, Reliability, Governability, and Bias Mitigation."[129]

NATO's risk mitigation responses include:

- NATO revised its AI strategy in July 2024, committing to accelerating the safe and responsible use of AI within NATO. The revised strategy addresses novel challenges, including

128 UK Cabinet Office (2024). Press release UK and its allies must stay one step ahead in new AI arms race. Available: https://www.gov.uk/government/news/uk-and-its-allies-must-stay-one-step-ahead-in-new-ai-arms-race

129 Werner, J (2024). NATO Updates AI Strategy to Address Emerging Challenges and Leverage Opportunities. Available: https://babl.ai/nato-updates-ai-strategy-to-address-emerging-challenges-and-leverage-opportunities/

AI-enabled disinformation, and reinforces existing principles. The strategy acknowledges the potential of synthetic biology to create novel organisms, whether for beneficial purposes, such as producing biofuels, or nefarious applications, including targeted biological weapons that evade attribution mechanisms.

- NATO established the Data and AI Review Board in NATO HQ under the Assistant Secretary-General for Innovation, Hybrid, and Cyber. The Board is mandated to provide guidance, set limits, and facilitate the exchange of information among allies. The board is developing "Responsible AI toolkits" and a "Responsible AI Certification Standard" to ensure that AI projects within the Alliance meet the NATO principles of responsible use.[130]
- To promote interoperability between AI systems and alliance-wide responsible adoption across the alliance, NATO is embedding an AI Testing, Evaluation, Verification & Validation (TEV&V) landscape.[131]

The People's Republic of China

Beijing has made significant efforts to regulate AI, and importantly, has made significant inroads in setting standards alongside creating rules. Chinese innovation in audit and disclosure related to AI, as well as the standards built into its AI-enabled technology exports, such as autonomous vehicles or digital tutors, is already globally significant.

130 Cheverton, D (2025). Atlantic Council Report: Second-order impacts of civil artificial intelligence regulation on defense: Why the national security community must engage. Available: https://www.atlanticcouncil.org/in-depth-research-reports/report/second-order-impacts-of-civil-artificial-intelligence-regulation-on-defense-why-the-national-security-community-must-engage/

131 Werner, J (2024). NATO Updates AI Strategy to Address Emerging Challenges and Leverage Opportunities. Available: https://babl.ai/nato-updates-ai-strategy-to-address-emerging-challenges-and-leverage-opportunities/

Similar to the EU, China's framework adopts a risk-based approach, although the categories and specific mitigations are applied differently across its various regulations. High-risk use cases, particularly those with "public opinion attributes or social mobilization capabilities," face enhanced control and oversight.

The PRC's risk mitigation responses include:

- In August 2023, China implemented the Interim Measures for Generative AI Services, which aim to govern publicly available generative AI services. The measures broadly prohibit generative AI models from producing content that "endangers national security and interests" or "harms the nation's image.[132]
- In September 2024, China's National Technical Committee 260 on Cybersecurity published its AI Safety Governance Framework V1, a risk-based approach to AI governance that ties risk levels to corresponding mitigations. The framework contains principles for governance, classifies potential risks, outlines mitigation measures, and provides safety guidelines for developers.[133] The framework explicitly states that sensitive data in high-risk fields such as CBRN "must not be used in the training" of AI models.[134]
- In February 2025, the China AI Safety and Development Association (CnAISDA) was established as a mechanism to

132 Baker & McKenzie (2023).China: New interim measures to regulate generative AI. Available: https://insightplus.bakermckenzie.com/bm/attachment_dw.action?attkey=FRbANEucS95NMLRN47z%2BeeOgEFCt8EGQ-JsWJiCH2WAWuU9AaVDeFglGa5oQkOMGl&nav=FRbANEucS95NML-RN47z%2BeeOgEFCt8EGQbuwypnpZjc4%3D&attdocparam=pB7HEsg%2FZ312B-k8OluOlH1c%2BY4beLEAezirm3%2BK7wMU%3D&fromContentView=1

133 DLA Piper (2024). China releases AI safety governance framework. Available: https://www.dlapiper.com/en/insights/publications/2024/09/china-releases-ai-safety-governance-framework

134 DLA Piper (2024). China releases AI safety governance framework. Available: https://www.dlapiper.com/en/insights/publications/2024/09/china-releases-ai-safety-governance-framework

represent China in international discussions on AI. The PRC government has described the institute as a counterpart to the AI safety institutes in Europe and the U.S.. China watchers have described the institute's emergence as a "significant victory for a group of policy entrepreneurs in China that has long warned about catastrophic AI risks".[135]

135 Singer et al. (2025) How Some of China's Top AI Thinkers Built Their Own AI Safety Institute. Available: https://carnegieendowment.org/research/2025/06/how-some-of-chinas-top-ai-thinkers-built-their-own-ai-safety-institute?lang=en



Source: Adobe Stock

II. Methodology

Initial Deepdive

Early in the project, the CSR team conducted a comprehensive review of the current state of biotechnology and artificial intelligence (AI). The team focused on how advances in biology and AI may make biological weapons easier to research, develop, produce, and use by nefarious actors. In particular, we paid attention to how the AIxBio convergence may accelerate the research and development of biological weapons as such tools become more accessible and capabilities like automation remove human error from life sciences research and manufacturing. The research has also drawn on recent technology risk mapping from key institutes such as The Centre for Long-Term Resilience, which in 2024 developed a distinct weapons development chain map focused on malicious actors intending to cause harm using a biological agent.

Using this framework, the team explored the current state of play of the risks and opportunities that arise at the AIxBio nexus. This also included a granular examination of currently available and specific AI-enabled tools and their potential impact on various phases of the biological weapons research, development, production, and weaponization process.

After mapping these capabilities and their implications for misuse, the team conducted a deep dive into the historical and current state of play of biological weapons from both state and non-state actors. The team considered it necessary to include, since capabilities alone do not explain what makes a state or non-state actor choose to pursue and use biological weapons; rather, these actors must also have reasons and a motivation to do so. In addition, to acknowledge the ties that exist

between advancing capabilities with AI, advances in the life sciences, and emergent capabilities in automation, we also chose to research and explore lab accidents as an angle to provide a more holistic understanding of the risks that may arise at the AIxBio nexus.

Finally, our team concluded with in-depth research on current initiatives and conversations that governments and other entities have led to better understand the current actions organizations and countries are taking to address the risks while harnessing the opportunities at this nexus. Many initiatives are currently focused on AI safety, with policies and implementations aimed at providing guardrails that prevent the malicious use of AI-enabled tools. In addition, many countries are exploring ways to integrate AI into other competitive activities, such as developing a robust international bioeconomy. Understanding who the leading actors and stakeholders are in this space will be key for this project as it progresses to the next steps.

Plausible Futures: Scenario Planning and Backcasting Exercises

While the United States government and others continue to improve their understanding of how AI will likely contribute to near-term biosecurity risks, technological innovations continue to outpace policy at an astounding rate. This dilemma restricts the maximization of opportunities at this nexus and the ability to future-proof policies that can effectively anticipate future risks.

This section describes AIxBio futures for the United States that CSR developed during two virtual workshops. Expert participants employed strategic foresight methodologies—a horizon scan, scenario planning

exercise, and backcasting exercise—to gain a deeper understanding of the plausible future trajectories for the AIxBio convergence nexus.

Strategic foresight is "an approach to long-term thinking and planning, often used to improve the ability to anticipate strategic opportunities and threats."[136] CSR employs methods within the strategic foresight toolkit, including scenario exercises, horizon scanning, and backcasting, to build a more nuanced understanding of how future uncertainties might unfold and stress-test expert assumptions and conceptualize global complexities that could enable or constrain alternative future pathways. We selected scenario planning exercises to support strategy development amidst uncertainty and challenge participants to map out, not predict, a range of alternative futures to better anticipate and prepare for the unknown. Using the 2x2 matrix method for scenario planning, the group developed four scenario narratives that articulate what the nexus would look like in each distinct future. These four scenarios are derived from the combination of two critical uncertainties – two factors or forces driving the future direction of the nexus, yet with significant uncertainties surrounding their influence and impacts on AIxBio, identified during the group's horizon scan.

As part of the initial phase of this project, the team conducted interviews with key experts, actors, and stakeholders across academia, industry, and government that highlighted:

1. key opportunities and risks arising from the AIxBio nexus;
2. the gap(s) that exist between the current state and predicted capabilities of AIxBio tools;
3. specific applications and tools that directly shift the risk landscape when it comes to the potential misuse of research and development of biological weapons;

136 UN Global Pulse (2023). Browse Foresight Tools. Available: https://foresight.unglobalpulse.net/how-tools/

4. considerations for mitigation measures to balance safety, security, and innovation; and
5. particular underappreciated or under-discussed aspects of the opportunities and risks at this convergence.

These interviews raised substantial differences in how experts identify and frame "risk" across communities involved at the nexus. For example, experts disagreed on whether AI may or may not contribute to biological weapons risks, as such a relationship would be variable: while AI tools could contribute toward biosecurity in some way, this contribution would be determined by the motivation, knowledge, and capability limitations with which they are employed, amidst other factors. Such differences are not necessarily disagreements, but are instead the result of present uncertainties surrounding the future trajectory of AIxBio. Thus, foresight tools were selected as the methods for identifying and organizing thoughts around critical uncertainties—those variables that would determine how AI intersects with biosecurity in the near term.

The interviews helped identify several current regulatory and geopolitical trends that are likely to play a critical role in shaping the future of this nexus by 2030. One issue is the challenge of balancing innovation and regulation: the recent revocation of former President Biden's Executive Order on "Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence" by the Trump Administration over concerns that it would hinder innovation is a stark contrast to the EU's AI Act that formalized a comprehensive legal framework for the safe, sustainable, and transparent development of AI systems–one

that highlights the critical future uncertainties of the domestic and international AI regulatory landscape.[137, 138]

Another core issue raised in interviews is how and to what degree biotechnology developments will accelerate the likelihood that state and non-state actors develop new or improve on existing WMD programs. AI could optimize and increase the efficiency of medical countermeasures for some actors while also enhancing bioweapons R&D processes for nefarious ones, creating both offensive and defensive gains that are difficult to weigh against each other. Likewise, the unknowns surrounding the state of geopolitical relations–such as the future of Russia's war against Ukraine, how tensions in the South China Sea will fare, and whether cooperation will continue between China, Russia, Iran and North Korea–question the stability of the future global security environment in which biosecurity and AI will intersect and evolve.[139]

With these issues in mind, participants first performed a short horizon scan of the different socio-political, technological, demographic, diplomatic, military, and economic factors that they believe will "drive" the future landscape of risks and opportunities at the AIxBio nexus by 2030.

137 European Commission (2025). The EU Artificial Intelligence Act. Available: https://artificialintelligenceact.eu/

138 The White House (2025). Initial Rescissions Of Harmful Executive Orders And Actions. Available: https://www.whitehouse.gov/presidential-actions/2025/01/initial-rescissions-of-harmful-executive-orders-and-actions/

139 U.S. Office of the Director of Intelligence (2025). Annual Threat Assessment of the U.S. Intelligence Community. Available: https://www.dni.gov/files/ODNI/documents/assessments/ATA-2025-Unclassified-Report.pdf

Drivers included:

- The state of biosurveillance capabilities, the level of politicalization of pandemic preparedness and response measures
- The state of research and development for AI automation
- The degree of commercial success of biotechnology
- whether governments will nationalize AI capabilities
- The level of respect governments have for sovereign borders during disputes
- AI accessibility, and
- whether international norms against the development of bioweapons remain.

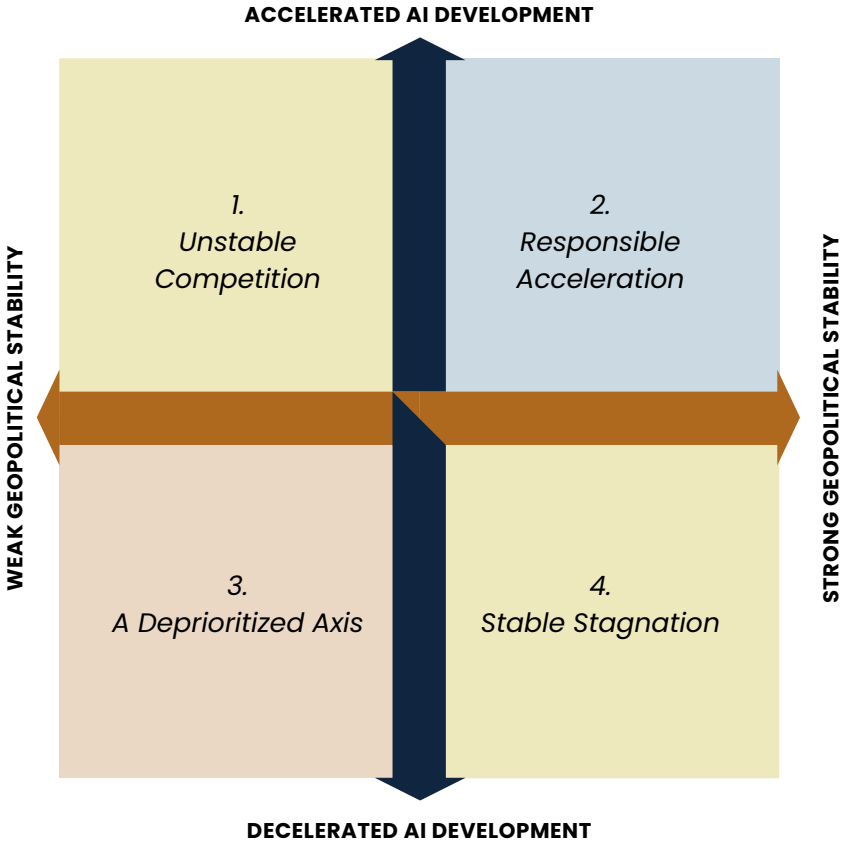
From this scan, the group then selected two drivers that would likely have the most substantial impact on this future, with uncertain outcomes: the speed of AI development and technological proliferation, and the level of geopolitical stability.

AI development and technological proliferation: This driver examines the rate of AI development over the next five years. Accelerated development could positively affect the nexus by enabling biotechnology developments, supporting growth within the bioeconomy, and increasing the accessibility of other AI-enabled opportunities. However, in the case of decelerated AI development, progress in innovating new AI systems slows down. What brings on this reduction in innovation, if it occurs, remains to be seen, but could include a stricter regulatory environment reinforced by states, a lack of government support for domestic AI companies and their growth, restrictions on resources and accessibility posed by the nationalization of AI systems, and more. A slowdown in AI growth could also indicate

a more sustainable innovation trajectory that, when coupled with developments in biosecurity and biotechnology, fosters a more stable AIxBio nexus.

Geopolitical situation: This driver considers the stability of the geopolitical environment in 2030 and its influence on pathways for the intersection of Bio and AI. An unstable geopolitical situation would increase tensions between states with AI capabilities. For example, competition might escalate due to the race to advance technologically and integrate new systems across military and defense capabilities. Geopolitical tensions could hinder the development of AI and AI-enabled biotechnology by creating an unstable regulatory environment that hinders their growth, or a disparate regulatory environment that fosters the proliferation of bioweapons programs and nefarious uses of new dual-use technologies. If geopolitical stability is high, on the other hand, some of these risks might be mitigated through improved cooperation among states over AI governance, augmented data sharing, cooperation over an accessible market structure for the growth of AI and bio-economies, and the promotion and protection of norms on bioweapons and mechanisms for improving transparency on compliance with the BWC.

Participants then articulated four alternative, yet plausible, future scenarios based on the combinations of different values–AI development either accelerates or decelerates, and geopolitical stability is either weak or strong–for the two key drivers:



SCENARIO 1: WEAK GEOPOLITICAL STABILITY/ACCELERATED AI DEVELOPMENT

In this scenario, although AI-first companies initially adopt new AI models, biotech companies also leverage AI to accelerate their research, development, and deployment, leading to rapid growth in the sector. AI-enabled biotech companies receive additional support from state sponsors as they become a critical sector for strategic competition.

Together, this leads to growth in the AIxBio sector and unfettered proliferation of dual-use R&D technologies among various state and nonstate actors. Bioweapons capabilities are added to the armamentarium of many state-sponsored terrorist groups.

However, it also leads to states investing in bioweapons capabilities. In the context of a weak international system, states utilize these capabilities in a deniable manner during a grey-zone conflict.

SCENARIO 2: STRONG GEOPOLITICAL STABILITY/ ACCELERATED AI DEVELOPMENT

In this scenario, governments cohere on a nuanced understanding of AI, viewing it as a strategic capability that can deliver significant competitive advantages. Yet, they are clear-eyed about the downstream risks.

AI-enabled biotechnology companies receive additional support from state sponsors, with regulatory conditions that serve as effective guardrails, incentivizing responsible innovation rather than stymying development.

A strong culture of public-private collaboration emerges with industry leaders closely involved in UN efforts to regulate these areas internationally. This leads to avant-garde international agreements that allow governments to develop streamlined regulation that broadly incentivises responsible innovation. Governments seek to regulate their domestic AI industries in line with international agreements, while supporting AI companies as economic 'national champions.'

SCENARIO 3: WEAK GEOPOLITICAL STABILITY/DECELERATED AI DEVELOPMENT

In this scenario, absent continued AI advancements, the AI-enabled biotech industry faces obstacles in the form of a lack of high-quality datasets that current models require. AIxBio companies also receive little to no additional support from state sponsors because the area is generally prioritized even less than AI.

The proliferation of dual-use AIxBio technologies among sophisticated state actors remains unfettered, as the sector is not viewed as a key axis for strategic competition. States invest more in biodefense and bioweapons programs as norms erode around bioweapons development and use.

SCENARIO 4: STRONG GEOPOLITICAL STABILITY/ DECELERATED AI DEVELOPMENT

In this scenario, a social consensus emerges, recognizing AI-enabled biotechnologies as a public good with substantial downstream risks. Governments regulate heavily and stymie innovation. Leaders pay lip service to the need for global AIxBio governance, but it is increasingly deprioritised due to technological stagnation.

AI-enabled biotechnology companies receive support from state sponsors for specific technologies with stringent regulatory conditions that limit innovation. Yet, without continued AI advancements, the AI-enabled biotech industry faces obstacles in the form of a lack of high-quality datasets that current models require.

Following the horizon scan and scenario exercise, experts convened a second time for a virtual backcasting exercise to help them understand what the "preferred" future looks like for the AIxBio nexus, and identify the steps that could be taken over the next five years to achieve that preferred scenario (articulated in the next section). Taking the preferred scenario as a given, backcasting challenges participants to explain the factors and events that would have led to or enabled that ideal. Backcasting is another foresight tool that aids strategic planning

by working backwards from a more desirable future to determine the ways and means required to attain it. Often, scenarios describe the worst possible case–from black swan events to the most risk-filled version of the future. We selected backcasting as a method to challenge participants to break out of solely focusing on the worst possible case and instead conceive of an ideal, yet plausible, case. The goal was not to reach consensus on every detail in the preferred future, but to instead put participants into the mindset that if such an outcome is a given, what would have needed to be changed, unlocked, and what policies would have been implemented for such a future to emerge. Backcasting exercises can help define milestones, resources, policies, and other recommendations for the changes needed to attain and maintain the preferred future.

Participants discussed the positive aspects of the four scenarios they developed during the first workshop to craft a fifth future pathway. This pathway articulates a 2030 where the convergence of AI and biosecurity was not defined by risk, but by opportunity. Experts were then asked to conceive of the potential actions, factors, events, changes (institutional, regulatory, political, economic, technological, etc.), and policies needed to facilitate that ideal. The group's preferred scenario describes a relatively stable geopolitical environment coupled with the sustained and responsible pace of AI development:

SCENARIO 5: GEOPOLITICAL STABILITY/SUSTAINABLE & RESPONSIBLE AI DEVELOPMENT

In this scenario, great power competition persists, yet world powers increasingly forgo competition for cooperation. Russia and Ukraine come to an agreement resolving the war in Ukraine. Russia focuses on rebuilding its economy, including global economic ties. NATO maintains the status quo and does not threaten to further expand along Russia's periphery. China and Taiwan experience a detente, and China is optimistic about eventually re-integrating Taiwan in the

long term through negotiation. US-China relations warm, leading to increasingly intertwined economies and cooperation in other areas. The US maintains a strong military deterrent in the region, disincentivizing Chinese military action against Taiwan. Multilateral organizations play a mediating role in successfully stabilizing these conflicts, thereby improving the UN's reputation and enhancing its soft power. The US, China, and Russia increasingly view the UN as an effective mechanism for resolving disagreements.

By 2030, governments will have catalyzed sustained and responsible AI development through sophisticated regulation and strong public-private collaboration. A global social consensus emerges that recognizes the need for responsible innovation in AI. This contributes to the UN becoming a productive and respected forum for establishing international governance of new AI capabilities. Governments share a nuanced understanding of AI, viewing it as a strategic capability that can deliver significant competitive advantages, yet they are clear-eyed about the downstream risks. Governments seek to regulate their domestic AI industries in line with UN agreements, while also supporting AI companies as economic 'national champions'. Frontier laboratories in the US, UK, EU, and PRC share a common understanding of the risks associated with AI, particularly in biosecurity, cybersecurity, and the integration of AI into safety-critical systems. A strong culture of public-private collaboration emerges with industry leaders closely involved in UN efforts to regulate these areas internationally. This leads to avant-garde international agreements that allow governments to develop streamlined regulation that broadly incentivises responsible innovation.

These factors facilitate responsible innovation, championed as AI-enabled biotechnology development accelerates. Although AI-first companies initially adopt new AI models, biotechnology companies also leverage AI to accelerate their research, development, and deployment, leading to rapid growth in the sector. A social consensus emerges, recognizing AI-enabled biotechnology as a public good that is revolutionizing drug discovery, improving global public health outcomes, and reducing the fiscal burden of healthcare. AI-enabled biotechnology companies receive additional support from state sponsors, with regulatory conditions that serve as effective guardrails, incentivizing responsible innovation rather than stymying development.



Source: Denys Kuvaiev / Adobe Stock

III. Policy Recommendations for the US Government

Findings from Expert Scenario Exercises & Key Stakeholder Engagement

The intersection of artificial intelligence and biotechnology (AIxBio) presents both tremendous opportunities and significant risks. A backcasting exercise conducted with our group of AIxBio experts (explained in the prior section) and follow-up engagement with key stakeholders yielded the following practical framework and tangible recommendations for the US government to achieve the best-case scenario for societal resilience to AIxBio risks over the next five years.

A. Strengthening the Biosecurity and Biosafety Foundation

With AI serving as a threat multiplier[140] to existing biorisks in both quantity and quality, developing and effectively implementing strong biosecurity and biosafety policies has become even more critical. Given the dual-use nature and democratization of both AI and biotechnology,

140 Goodman, Sherri, and Pauline Baudu. "BRIEFER: Climate Change as a 'Threat Multiplier': History, Uses and Future of the Concept."January 2023. Available at: https://climateandsecurity.org/2023/01/briefer-climate-change-as-a-threat-multiplier-history-uses-and-future-of-the-concept/.

regulating or standardizing the relevant AI development and deployment will not suffice to mitigate the biorisks associated with the rapid development of foundation models and AI-powered biological tools. It follows that a comprehensive framework for societal resilience to AIxBio risks also includes a strengthened foundation of biosecurity and biosafety policies already in practice and advancing.

To counter both state and non-state actor threats, the US government needs to continue advancing an even more **comprehensive deterrence-by-denial strategy** by establishing increased national and international preparedness, early warning systems, and response capabilities. With these measures in place and implemented well, biological incidents would cause much less damage and destruction in terms of lives and the economy. In the case of deliberate misuse of biological agents, this could effectively reduce the rationale for them being pursued and used as weapons. The United States government is already working towards this vision, and it parallels expansive thinking about deterrence methods in the modern era across the full range of strategic threats. It is critical to sufficiently fund targeted capability development and maintain the relevant subject-matter expertise at the Department of Defense, Department of State, Department of Health and Human Services, and Centers for Disease Control and Prevention, and elsewhere. Additionally, the US government must maintain and invest in close collaboration and coordination with allied and partner countries whose capacities to detect and respond to biological threats—given their transnational nature—directly protect American lives, territory, and security interests.

To counter non-state actor threats, the US government must rapidly adopt common-sense biosecurity policies such as **mandatory DNA synthesis screening** and **data security frameworks**, critical in addressing bioweapons materials and knowledge proliferation, respectively. On the materials side, effective regulation must also consider

the potential evasion of screening protocols through foreign suppliers or decentralized benchtop DNA synthesizers. Following the model of anti-money laundering regulations, the US government should require comprehensive customer verification requirements for DNA synthesis companies and device manufacturers, both domestic and international, with access to the US market. This includes identity verification, end-use monitoring, and suspicious activity reporting. These steps have been explored deeply for years by a range of expert groups, and are primed for implementation.[141]

The US government should also adopt frameworks for controlling and monitoring access to biological data with dual-use potential, allowing timely detection, tracking, and response to malicious or highly dangerous activities. A centralized database requiring account registration and ID verification could offer a balanced approach that protects national security interests while democratizing data.

At the same time, it is crucial to **update biosafety frameworks** to keep pace with the risk and magnitude of accidental lab leaks, which are increasing due to enhanced AI applications. For instance, expanding the capabilities of synthetic genomic technologies that enable intrinsic genetic containment and bio-cryptography may warrant updates to applicable containment strategies, dictating that such systems be required by default, especially in high-risk research such as synthetic organisms used in open environments. In a beneficial application of AI in biosafety, AI-assisted monitoring systems could help detect potential safety breaches before they result in laboratory accidents or releases.

141 Organizations working in this space include: The International Biosecurity and Biosafety Initiative for Science (IBBIS), SecureDNA, Engineering Biology Research Consortium (EBRC), Georgetown's Center for Security and Emerging Technology (CSET), The Johns Hopkins Center for Health Security (CHS).

Generally, **biosafety regulations must be better tailored to research risk and benefits.** For instance, some extremely high-risk, potentially AI-enabled, research involving biological material or manufacturing would likely warrant at least a moratorium. At the same time, the risks and benefits are further explored. An example would be research into creating mirror life: lifeforms composed entirely of mirror-image biological molecules, DNA and RNA made from "left-handed" nucleotides, and proteins from "right-handed" amino acids. While potentially still a decade and large investments and technical advances away, an interdisciplinary group of 38 researchers published a report warning of serious risks in 2024, such as the next pandemic from potential mirror life pathogens circumventing the human immune system and irreversible ecosystem disruption from mirror life outcompeting existing organisms.[142] Following the reports' publication, the expert community started gathering more broadly, eliciting emerging agreement that research in this space should simply not occur.[143] In such governance, it is important to differentiate mirror life from mirror molecules, which carry a different risk-benefit profile (potentially significant benefits with no pandemic risks, as the molecules created are not living, replicating organisms). There are therapies already being developed using mirror proteins' reversed chirality for their high degradation resistance—effectively extending the life of a protein-based drug. The discourse in this field is a strong example of just how tailored and nuanced many policies will need to be.

Similarly, prion research that focuses on the study of neurodegenerative diseases carries a similar risk-benefit profile that regulation would have to carefully consider. While prions' unique characteristics of extreme resistance to conventional decontamination methods and ability to cause fatal infections and disease make them an incredible threat if

142 Adamala, Katarzyna P., et al. "Confronting risks of mirror life." *Science* 386.6728 (2024): 1351-1353, http://doi.org/10.1126/science.ads9158.

143 Adamala, Katarzyna P., et al. "Confronting risks of mirror life." *Science* 386.6728 (2024): 1351-1353, http://doi.org/10.1126/science.ads9158.

they were to be used as a biological weapon, a potential benefit of their further research could be to discover a therapeutic treatment or cure against prion diseases. In this case, short-term policies facilitating safely-performed work and limiting the kinds of synthetic prions that can be created could lead to the long-term benefit of finally developing a treatment for infection. Given the benefits of such research, a simple ban would be much less appropriate, although a time-constrained halt of such research could facilitate targeted and enforceable regulatory development.

As of this writing, the US government is considering significant restructuring of policies and processes to streamline how these risk-benefit calculations are carried out and treat such unique and consequential issues with the nuance they warrant. Among these considerations is the creation of an independent agency focused on biosafety and biosecurity in life sciences research.[144]

Following the One Health approach, whose goal is to achieve "optimal health outcomes recognizing the interconnection between people, animals, plants, and their shared environment,"[145] these strategies and associated regulatory actions will require **a significant increase in the collective willingness and coordination between a breadth of governmental and non-governmental institutions** to guard against biorisks. Indeed, some US biosecurity policy experts are beginning to consider whether grander organizational structure questions should be asked to make such coordination stronger, but also significantly

144 Snyder, B. C., et al. Envisioning an Independent Bioresponsibility Authority to Safeguard U.S. Leadership in the Life Sciences. Scowcroft Institute of International Affairs, Bush School of Government & Public Service, Texas A&M University, January 2025, https://bush.tamu.edu/wp-content/uploads/2025/01/Envisioning-an-Independent-Bioresponsibility-Authority-Jan-2025-2.pdf.

145 Centers for Disease Control and Prevention. (2025). *About One Health*. U.S. Department of Health and Human Services. https://www.cdc.gov/one-health/about/index.html

faster and more agile, given how dynamically the AI and bio fields are changing, in themselves and in combination.

B. AIxBio Policy

The Trump administration's new AI Action Plan, released on July 23, 2025, reinforces CAISI's role to evaluate frontier AI systems for national security risks in partnership with frontier AI developers, in collaboration with other agencies with relevant expertise in CBRNE and cyber risks. The Trump administration also recognized the importance of the AIxBio risks when it tasked the rebranded Center for AI Standards and Innovation (CAISI) to focus on "demonstrable risks, such as cybersecurity, biosecurity, and chemical weapons" for establishing voluntary agreements with private sector AI developers and evaluators, and leading unclassified evaluations of AI capabilities that may pose risks to national security. This coordination with the private sector is critical given the industry-driven nature of this technology development and AI companies' resources, insights, and expertise. The continuity in this area of US policy is important and commendable.

CAISI's voluntary agreements with the private sector, as well as AI companies' voluntary participation in the Frontier Model Forum and development of the EU General-Purpose AI Code of Practice, are already allowing best practices like red teaming, testing, and validation to become industry standards. However, with shareholder interests and economic competition creating uneven incentives for AI companies, voluntary standards and commitments are unlikely to suffice in the long term. If the Trump administration is not proactive enough in providing a **national regulatory framework**, companies might face a convoluted regulatory landscape with contradictory or

onerous-to-implement measures. Such an environment would be particularly difficult to navigate for start-up and early-growth businesses that don't have established compliance teams.

Most importantly, the AI Action Plan gives direction to prioritize the recruitment of leading AI researchers at Federal agencies, including NIST and CAISI within DOC, DOE, DOD, and the IC, to ensure that the Federal government can continue to offer cutting-edge evaluations and analysis of AI systems. In addition to sufficiently funding CAISI and other departments and agencies here, the U.S. government should increase direct hiring authorities and leverage flexible **hiring authorities** such as the Intergovernmental Personnel Agreement Act that can be especially helpful for short-term efforts to infuse private sector talent into the government. To effectively fulfill its mandate, CAISI would likely also benefit from more flexible and targeted **contracting vehicles** that fit its fast-paced mission, growing industry stakeholders, and unique subject-matter expertise.

Additionally, given the sensitive subject matter and lack of relevant efforts at other US government agencies, the administration should increase CAISI's mandate to include **classified evaluations** of AI capabilities. This should include fast-tracking the security clearance process for new hires, aligning with AI as one of this administration's key priorities.

Moreover, the Trump administration should develop **guidance for dual-use relevant AIxBio research** and coordinate more closely with relevant researchers, all while protecting beneficial innovation. The federal government should convene regularly to both monitor their breakthroughs and see collaborative ways it can help researchers verify their research is not misused. For a start, AI companies, as a **condition of receiving federal funding**, must adhere to federal guidance if their product has the potential for AIxBio dual-use.

Additionally, the federal government should establish **intelligence back channels** between researchers working on applying AI tools to biological data, both in industry and academia, and intelligence agencies (for example, the FBI and DHS). Such a mechanism should allow and encourage academics to report any suspicious activity and employees in industry to whistleblow if they see best practices being ignored. By creating incentive structures for AI companies and academics, intelligence agencies can be informed of any misuse indicators, theft, or attempted theft of American scientific work. Additionally, the US could look to the UK's Secure Innovation, a Five Eyes intelligence partnership campaign, where small and mid-size tech companies can receive security reviews to prevent theft of their intellectual property by state and state-backed competitors and cyber criminals at a government-subsidized rate.[146]

C. AIxBio Technical Governance

While AIxBio technical governance has quickly developed on a largely voluntary basis over the last couple of years, the rapid technical developments and applications at the AIxBio intersection require sustained investment in qualitatively updating and quantitatively universalizing the implementation of such standards—for both foundation models and AI-powered biological tools.

146 https://www.npsa.gov.uk/specialised-guidance/secure-innovation

General-purpose AI: Foundation Models

As the Frontier Model Forum describes in its first technical report, evaluations can vary significantly in both resource intensity and strength of evidence. **Automated evaluations** like scientific knowledge benchmarks are quicker and easier, but less thorough, than more **comprehensive evaluations** such as expert-led red teaming and human uplift studies.[147] The US government needs to make clear: When evaluating for national security-relevant capabilities such as bioweapons development or autonomous biorisks, we cannot take any shortcuts.

As a start, CAISI (at the time still AISI) released the second public draft of its guidelines on Managing Misuse Risk for Dual-Use Foundation Models (NIST AI 800-1), which outlines **voluntary best practices** for identifying, measuring, and mitigating risks to public safety and national security across the AI lifecycle. They follow a simple framework that the US government should **base broader regulatory requirements on**:

1. Identify potential misuse risk,
2. Plan to manage misuse risk,
3. Protect the model from unauthorized access (if necessary),
4. Measure the misuse risk associated with model deployments,
5. Mitigate misuse risk before deployments,
6. Monitor and respond to misuse, and
7. Disclose misuse risk management practices.[148]

147 Frontier Model Forum Technical Report, "Frontier Capability Assessments," April 22, 2025, https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf.

148 Managing Misuse Risk for Dual-Use Foundation Models, (2025). U.S. AI Safety Institute, https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.800-1.ipd2.pdf.

To implement these requirements, which are general and high-level enough for potential regulatory language, AI developers could look at NIST's upcoming final draft of these guidelines that can be more easily updated to keep pace with technical developments.

Existing **industry best practices** in line with objective 4 (measure misuse risk associated with model deployments) include a combination of relative capability, bottleneck, and threat simulation assessments.[149] While most of these are common among the top industry leaders, adoption still varies and hinges on resources and implementation.

Relative Capability Assessments compare the capabilities of new models against those of previously evaluated models to make inferences about relative risk, using standardized tests called benchmarks. For biological capabilities, Language Agent Biology Benchmark (LAB-Bench) measures performance on practical biology research tasks, while the Weapons of Mass Destruction Proxy (WMDP) benchmark includes questions that assess knowledge that may be related to what malicious actors may be interested in biosecurity, cybersecurity, and chemical security.[150] Frontier model companies have assessed these currently best test proxy capabilities relevant to bio threat models.[151]

Bottleneck Assessments test whether models possess specific capabilities that domain experts believe would remove "bottlenecks" to severe real-world harm.[152] It is crucial to test models with access

149 Frontier Model Forum Technical Report, "Frontier Capability Assessments," April 22, 2025, https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf.

150 "The WMDP Benchmark: Measuring and Reducing Malicious Use With Unlearning," https://arxiv.org/abs/2407.10362; https://www.wmdp.ai/.

151 Frontier Model Forum Technical Report, "Frontier Capability Assessments," April 22, 2025, https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf.

152 Frontier Model Forum Technical Report, "Frontier Capability Assessments," April 22, 2025, https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf.

to all tools they might be given access to, such as the internet. As the Frontier Model Forum report describes, "a model with access to the internet and other tools will perform differently on the same Bottleneck Assessment compared to a model without access to these tools." However, it appears impossible to account for changes within those tools, such as new dual-use biological data being uploaded to the internet, that are outside of the control of frontier model companies. External factors and larger contexts generally present challenges for this type of assessment.

Threat Simulation Assessments involve "simulating" substantial segments of threat scenarios from end to end to estimate more directly the extent to which the model would enable these scenarios.[153] For instance, some simulations have compared the outputs of biology novices with AI assistance with those of a control group using only standard online resources when completing technical tasks relevant to pathogen engineering, and similar types of assessments have been conducted with experts.

Each of these types of assessments has limitations, thus currently requiring their **combination and continuous updating** to be considered best practice. With the right set of experts, training, and partnerships, thorough evaluations would not necessarily slow down innovation but rather allow for continued testing throughout model development, from early-stage to fine-tuned models. This is critical as evaluations might not reveal all necessary information on a model's bioweapons and other national security risks until later in a development state or even throughout deployment.

Finally, to avoid regulatory gaps, the US government should also **require downstream actors to implement safeguards** such as more

153 Frontier Model Forum Technical Report, "Frontier Capability Assessments," April 22, 2025, https://www.frontiermodelforum.org/uploads/2025/04/FMF-PDF-Frontier-Capability-Assessments_-Technical-Report.pdf.

context-specific assessments when adapting models and integrating them into high-risk applications with potential new deployment contexts and use cases. This would, for instance, be the case when integrating them into biological tools that could, in combination, elicit new and unforeseen biological risks.

Narrow AI: AI-powered biological tools

While the US government and most external studies have so far focused on foundation models, narrow AI-powered biological tools require an increased focus, starting with government and industry research and evaluations.[154] For instance, CAISI, which is more directly tasked with evaluating foundation models, should also establish **dedicated workstreams for evaluating biological design tools** (BDTs). While not explicitly spelled out in the recent AI Action Plan, this arguably falls within the line of effort to "Build, maintain, and update as necessary national security-related AI evaluations through collaboration between CAISI at DOC, national security agencies, and relevant research institutions."

AI can augment current nucleic acid screening methods. Some studies suggest AI-enabled BDTs can be used to evade traditional sequence-based biosecurity screening methods;[155] research needs to be conducted to advance screening techniques that possibly themselves leverage AI/ML in adding to sequences of concern. The International Gene Synthesis Consortium (IGSC) is a trade organization in which industry-leading gene synthesis companies share screening protocols and best practices, collaborating extensively with public sector entities such as governments and international organizations.

154 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, page 28.

155 Bruce J. Wittmann, et. al., "Toward AI-Resilient Screening of Nucleic Acid Synthesis Orders: Process, Results, and Recommendations." bioRxiv, December 4, 2024, https://www.biorxiv.org/content/10.1101/2024.12.02.626439v1.

This continued cooperation with government bodies is paramount in enabling enforcement mechanisms. Still, this enforcement would also need to apply to synthesis companies that do not adhere to best practices and may not be members of IGSC. Thus, in addition to research into biosafety best practices, the federal government is best equipped to develop infrastructure to facilitate data sharing between nucleic acid synthesis providers. This would help prevent security gaps and ensure that synthesis companies aren't disadvantaged by the costs of taking on this security measure independently. The 2025 AI Action Plan[156] requires this screening and information sharing to be conducted, creating enforcement mechanisms for researchers receiving federal funding, and it also mandates collaboration between CAISI and research institutions to perform national security-related AI evaluations.

The Center for AI Standards and Innovation should **develop an if/then strategy for the BDT capability threshold** that would trigger a federal security review and a specified countermeasure.[157] To facilitate what capability thresholds among BDTs would trigger concern, the federal government could convene academic researchers and industry partners to develop voluntary standards and best practices. Because this strategy suggests that, if a certain capability emerges, it must be coupled with a specific countermeasure, it would require case-specific security evaluations and reviews before a model can be released, necessitating a partnership with BDT companies. While such an approach may work in the short term, with scenarios that seem possible to develop given today's AI, it will be difficult in the long term as AI improves and becomes less predictable. Some examples of significant capability uplift of AI-enabled biological tools that the National Academies

156 Winning the Race: America's AI Action Plan, (2025), https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf.

157 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, pages 6-7.

report flagged as concerning when they emerge include: (1) models that can use AI to design infectious agents capable of fully replicating, (2) models that predict transmissibility and pathogenesis with high accuracy, (3) computational tools (like virtual cells) so accurate that predictions made using AI models (in silico) are reliable enough that they no longer require wet lab (in vitro or in vivo) testing testing for confirmation, or (4) AI-agent driven automated laboratories.[158]

In preparation for a future in which Artificial General Intelligence (AGI) interfaces directly with BDTs and bioengineering machinery, the US should **invest in research around AI model alignment and cybersecurity best practices** for autonomous and cloud-based laboratories. As biomanufacturing leverages AI and robotics to increase in scale and capability, the potential methods for attack and the risks associated with those attacks also increase. This will necessitate "increasingly strong deployment and security protections."[159] As more labs move to automate scientific experimentation, methods must be put in place to keep these autonomous tools secure, prevent attacks, and automatically monitor what chemical or biological material is being produced in a privacy-preserving manner. The federal government needs to **monitor the use of autonomous labs and their cloud security practices**, allowing for incident reporting and requiring these labs to undergo regular adversarial testing. The digital-to-physical boundary currently safeguards against the undetected production of bioweapons[160]–but this may not remain the case if biomanufacturing moves away from having a persistent human-in-the-loop. Because AGI loss of control in biomanufacturing could pose an extreme risk, federally funded research into best practice protocols in the case of a loss of

158 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, page 37.

159 Anthropic. "Activating AI Safety Level 3 Protections," May 22, 2025, https://www.anthropic.com/news/activating-asl3-protections.

160 https://helena.org/projects/helena-biosecurity

control needs to be conducted; additionally, CAISI should prioritize alignment research (research to ensure advanced AI systems behave as intended and remain controllable), and lines of communication should be established between autonomous labs and federal law enforcement to quickly alert federal authorities in the event of an attack.

Data Policies around AI Development

Develop data best practices to enable the collection of high-quality data, and research adherence to current data provenance standards for biological data. Experts constantly discuss the importance of high-quality biological data in the development of powerful AI tools, suggesting we should treat this data like a strategic asset and also safeguard it as such.[161] By requiring good data practices, high-quality data collection and sharing, and quality data provenance and documentation practices, the aim can be the creation of better quality biological data and subsequently better quality BDTs. Data provenance is the historical record of data source(s) and the changes made to the data throughout its lifecycle. Compared to the opaque data provenance practices used in the development of foundation LLM models, in theory, there are standardized procedures dictating provenance norms for biological material and data.[162] Yet, compliance with these standards is not well understood or monitored, suggesting that BDTs may have been or could be trained on a variety of sensitive biological data that could be easily misused. These issues are widely discussed across the health and biomedical research fields, in addition to efforts focused on the AIxBio nexus, and will take great detail to address.

161 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, pages 8, 65, 72.

162 International Organization for Standardization. (2023). "Biotechnology—Provenance information model for biological material and data" (ISO/TS 23494-1:2023).

The US intelligence community should **perform horizon scans of current data being collected and aggregated with biological data** to gauge what capabilities future AI tools may possess.

"[T]he availability of high-quality, robust data can be the leading indicator of an emerging or developing AI model capability," according to the National Academies' 2025 report, "The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations."[163] This means that the kinds of data being collected today directly inform the models of the future. Moreover, AI models in research and development today can form the backbone of future technologies, which may be especially relevant for dual-use technologies. For example, a recent paper[164] found a direct link between academic AI publications about computer vision and subsequent developments in surveillance technology. With Evo2 as an example, the aggregation of RNA, DNA, and protein data with genomes of prokaryotes and eukaryotes labelled with functional information could serve as a leading indicator for the development of an AI capable of multi-domain modeling and novel DNA sequence generation between organisms. Similarly, an entity known to be aggregating human genetic sequences, human sequences known to increase vulnerability to viral infection, and ways viruses can be engineered to be more infectious to targeted populations, may be seen as a leading indicator of an entity that may soon be able to create precise bioweapons. Regulating or attempting governance of AI models will always entail **managing unpredictable changes**, whereas governance of data collection and aggregation can enable the creation of more prescient policy.

163 National Academies of Sciences, Engineering, and Medicine. 2025. The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, page 58.

164 Kalluri, Pratyusha Ria, et al. "Computer-vision research powers surveillance technology." Nature (2025): 1-7.

AI for Early Threat Recognition and Response

AI should be leveraged for use in biosurveillance and biological countermeasures. Federal public health agencies should ensure that any AI tools intended for biosurveillance are first stress-tested and subject to regular penetration testing, and the details of data provenance and data robustness should be verified.[165] In the absence of this kind of training and assurance of high data standards, a bad actor could inject malicious data into tools or exploit back doors that may be present in public datasets, leading to delays of needed countermeasures.[166] Experts have suggested the US establish a public-private partnership akin to Operation Warp Speed, which accelerated "the development, manufacturing, and distribution of COVID-19 countermeasures."[167, 168] Still, this partnership would specifically focus on AIxBio tools to be rapidly developed and launched for pandemic response[169] including AI/ML to aid in vaccine design and efficacy modeling[170]—leveraging virtual cells to observe the functionality of novelly designed proteins and biomolecules to understand immune response. Additionally, AI/

165 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, pages 55-56, 79.

166 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, pages 56-57, 79-80.

167 U.S. Department of Defense. (2020, May 15). Trump Administration Announces Framework and Leadership for 'Operation Warp Speed'. Retrieved from https://www.defense.gov/News/Releases/Release/Article/2310750/trump-administration-announces-framework-and-leadership-for-operation-warp-speed/.

168 Explaining Operation Warp Speed, https://health.mo.gov/living/healthcondiseases/communicable/novel-coronavirus-lpha/pdf/fact-sheet-operation-warp-speed.pdf.

169 National Academies of Sciences, Engineering, and Medicine. (2025). The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations. Washington, DC: The National Academies Press. https://doi.org/10.17226/28868, pages 5, 57.

170 Modeling how antibody responses may determine the efficacy of COVID-19 vaccines, February 28, 2022, https://www.nature.com/articles/s43588-022-00198-0.

ML can be leveraged for in silico testing of different populations and demographics to aid in determining optimal vaccine schedules.[171]

D. AIxBio Diplomacy

With the global nature of AIxBio risks, there is an urgent need to strengthen the United States' **multilateral capacity to address technological change**, beyond merely enhancing awareness of the implications of emerging technologies for the production and use of biological weapons. For the United States, this means sufficiently staffing relevant Department of State offices and their interagency counterparts, and utilizing flexible hiring mechanisms for specialized subject-matter experts. While multilateral consensus venues like the Biological Weapons Convention or a new legally-binding treaty would present practical challenges given the urgency and rapid technical developments of the AIxBio risk problem, other avenues like UN General Assembly resolutions or more informal coalitions of the willing could help establish international norms to safeguard against AIxBio risks.

Additionally, on the more technical risk mitigation side that CAISI leads for the United States, the most practical next step includes strengthening and globally expanding the **International Network of AI Safety Institutes,** which builds on the Seoul Statement of Intent toward International Cooperation on AI Safety Science from May 2024. Although the Network is not a standard-setting body, the United States could leverage its chair position to create international AI and data norms (data safeguards, clear consent, data control, and privacy-preserving tech) that will have implications for biosecurity.

171 A modeling-based approach to optimize COVID-19 vaccine dosing schedules for improved protection, May 25, 2023, https://insight.jci.org/articles/view/169860.

To maximize the beneficial uses of AI for biotechnology, the United States should explore leveraging more **strategic partnerships for the responsible transfer of AIxbio technologies** among trusted partners, for instance, under AUKUS Pillar II. Such a partnership could include streamlining export controls in line with international nonproliferation obligations. Collaboration on legitimate research and innovation should be contingent on and in recognition of each partner's adoption of adequate risk mitigation and technical oversight measures and capacities, ideally in coordination with the International Network of AI Safety Institutes.

Beyond the coalitions of the willing, the challenge will be getting key competitors like China to adopt similarly high standards. Several AI track 2 dialogues between the US and China, if effectively coordinated and with the level of political support on both sides, could be leveraged to transition into a **track 1.5 or 1 dialogue with China** to exchange best practices for mitigating the less political non-state actor risks arising out of the AIxBio nexus. This common underlying interest could have the positive secondary effect of opening dialogue and finding pathways for coordination in a time otherwise dominated by heated competition. As the United States' most advanced and main competitor in this space, it will be crucial to identify and leverage **alternative channels for engagement** and cooperative risk reduction with China. Another channel could involve the United Kingdom, as a de facto mediator and leader in the space with its flagship UK AI Safety Institute, driving AI security talks with China.

On the flip side, the United States needs to continue carefully evaluating **tailored export controls** for countries for which it has concerns about compliance with the Biological Weapons Convention. While chip export controls have become politicized and mixed with economic interests, successful US competition will require continued case-by-case

assessment, considering our adversaries' supply-chain workarounds and indigenous production capabilities.

E. Conclusion

By strengthening the biosecurity and biosafety foundation, building on voluntary commitments with a national AIxBio regulatory framework, and advancing AIxBio technical governance, the United States can build resilience against AI-enhanced biological threats while preserving the benefits of technological innovation. Success will require sustained funding and commitment from governments, the private sector, and the broader research community.



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